



Predictive transmission accuracy modeling of RV reducers under geometric errors and wear effects

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Received: 18 August 2025 / Accepted: 3 February 2026

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Abstract

This study proposes a novel predictive transmission accuracy model for RV reducers, integrating geometric errors and tooth wear effects through a methodology combining Archard's wear theory with a 20-degree-of-freedom quasi-static formulation. The approach employs mass-spring equivalence to decompose component misalignments into equivalent spring deformations, while tooth contact analysis and loaded tooth contact analysis dynamically resolve time-varying contact forces, sliding distances, and pressure-dependent wear coefficients. Critical factors including tooth profile modifications, eccentricity errors, and wear-induced geometric deviations are incorporated through iterative geometry recalibration. Analytical procedures quantify manufacturing error propagation paths and wear progression in cycloid-pin gear pairs using discretized surface meshing and adaptive threshold-based parameter updates. Validation methodologies encompass RecurDyn dynamic simulations for force distribution verification and experimental prototype testing for transmission error precision measurement. Results demonstrate that the model effectively captures error sensitivity dominance in cycloid transmission stages and wear-induced accuracy deterioration mechanisms, providing a foundational framework for precision design optimization of RV reducers under coupled error-wear interactions.

Keywords RV reducers · Transmission accuracy · Geometric errors · Tooth wear · Predictive modeling

Abbreviations

CMM	Coordinate Measuring Machine
E/E^*	Elastic modulus/ Equivalent elastic modulus
ν	Poisson ratios
r^*	Relative curvature radius
a	Pin distribution radius
r_r	Pin Radius
e	Eccentric distance
P	Pitch point
B	Cycloid gear width
k_i	Hertzian contact stiffness
z_l	Pin number

z_2	Tooth number of the cycloid gear
n	Number of crankshafts
$\mathbf{r}_f^{(1)}, \mathbf{r}_f^{(2)}$	The position vectors of the contact point on the cycloid gear 1 and pins 2
$\mathbf{n}_f^{(1)}, \mathbf{n}_f^{(2)}$	The unit normal vectors at the contact point
k	Wear coefficient
V	Wear volume of the material
S	Relative sliding distance
K	Dimensionless wear coefficient
P_r	Contact normal pressure
W	Material hardness
δ_i	Magnitude of the contact deformation
σ_{Hi}	Hertzian contact stress
T_d	External load
μ_d	Sliding friction coefficient
l_p, l_{ui}	Moment arms of F_i and F_{ui}
α_i	Micro angular displacement
r_{db}	Base circle radius
F_{ui}	Friction force of the i-th cycloid-pin pair
$\Delta\varphi_c$	Output angle variation
T_f	Frictional torque component

Technical Editor: Guilherme Rosa Franzini.

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F_i	Force of the i -th cycloid-pin pair
ϕ_{bli}	Backlash of the i -th cycloid-pin pair
l_{pi}	Distance between the contact point and pitch point P
r_{bg}	Distribution radius of the crankshafts on the cycloid gear
ψ_r	Distribution angle of adjacent pins
a_{Hi}	Half contact width
TCA	Tooth Contact Analysis
LTCA	Loaded Tooth Contact Analysis
DOF	Degrees of Freedom

1 Introduction

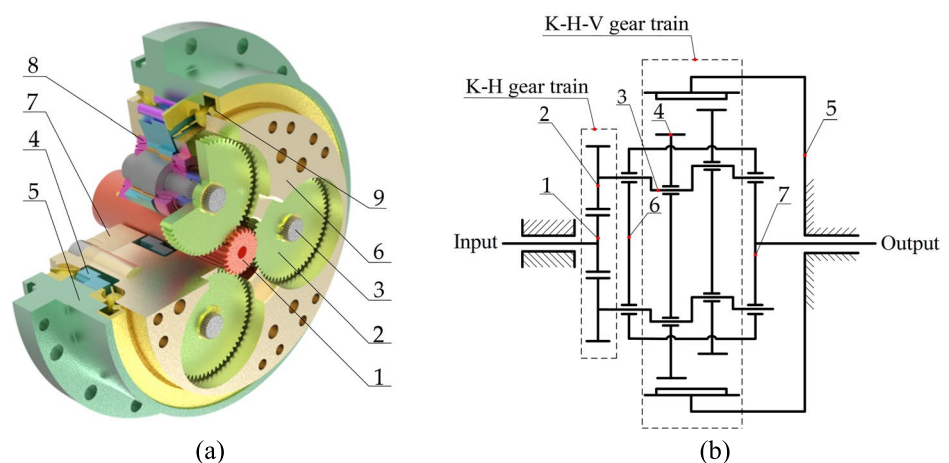
RV reducers have been widely employed in assorted industries, offering advantages over other reducers, like a compact structure, the ability to obtain a large reduction ratio, and high load capacity. As depicted in Fig. 1, which show a 3-D diagram and structural schematic, respectively. A classic RV reducer's structure consists of two main reduction stages: the initial reduction employs a planetary gear drive mechanism comprising a sun gear and typically three planet gears, while the subsequent reduction involves a cycloid-pin gear drive mechanism consisting of typically three crankshafts, two cycloid gears, a fixed pinwheel, and an output shaft. Analyzing the profile wear of cycloid-pin pairs is crucial for maintaining and enhancing the transmission performance of RV reducers.

Recent years have seen sparse research and literature on profile wear in cycloid-pin pairs, far less than that on spur, bevel, and helical gears. However, analytical methods applied to the latter can be extended to cycloid-pin pair wear analysis. The 1953-proposed Archard wear model remains the most critical and widely used in wear studies [1]. Using the Archard model, Flodin et al. [2] developed a wear prediction model for external spur gears with simulation results, while Tunalioglu et al. [3] established a theoretical

model for internal spur gears validated experimentally. Ren et al. [4] proposed a dynamic wear prediction model to explore the quantitative link between dynamic characteristics and surface wear, and Kahraman [5] combined torsional dynamic and surface wear models to study their interactions under linear/nonlinear conditions in planetary gears. For helical gears, Flodin et al. [6, 7] developed a wear prediction model and its simplified version, comparing their simulation results. Osman et al. [8] introduced an innovative model integrating surface wear in wide-faced solid spur and helical gears under quasi-static and dynamic conditions. Kahraman et al. [9] further studied how tooth profile deviations affect helical gear wear via a dedicated model. Park et al. [10] integrated loaded tooth contact analysis with the Archard model to create an efficient method for calculating hypoid gear tooth wear. Wang et al. [11] proposed a high-precision wear model for spur gears with rough surfaces, validated by comparing smooth and rough surface simulations. Pei et al. [12] predicted spiral bevel gear wear by considering surface roughness, mixed lubrication, friction, flash temperature, and dynamics, using TCA to obtain time-varying meshing parameters. In cycloid-pin research, Li et al. [13] developed a wear model based on the Archard model and Hertzian contact theory, accounting for tooth profile modification. Xu et al. [14] derived the sliding ratio equation for a novel spur cycloid drive and analyzed parameter influences on this ratio.

On the other hand, in light of analysis on the RV reducer system, attention has been attracted over the past decades. Blanche and Yang et al. [15, 16] employed pure geometric methods to develop a contact model for single-stage cycloid transmissions, enabling rotation error calculation under contact deviations. According to TCA, Li et al. [17, 18] proposed a numerical model for backlash-included cycloid-pin meshing, computing transmission error, mesh stiffness, and reduction vibration. Xie, Xu et al. [19–22] integrated multi-body dynamics and elastic contact theory to construct a multi-tooth mesh model, precisely determining contact tooth

Fig. 1 Structure of RV reducer: **a** 3-D diagram and **b** structural schematic diagram



count, moment rigidity, rotation precision, and tooth forces. Hidaka et al. [23] innovatively developed a static transmission precision model via mass-spring equivalence, decomposing component errors into respective action lines. This model efficiently calculates two-stage RV reducer (with two cycloid gears and three cranks) precision while accounting for single/component error combinations. Li et al. [24] built an RV reducer dynamic model to explore tooth modification and system errors on transmission characteristics. Zheng et al. [25] took BX-40E as an example, studying dynamic wear evolution's impact on RV reducer performance degradation using Archard theory. Song et al. [26] proposed a creative torsional stiffness calculation method for RV reducers based on Hertzian contact, considering tooth modification and variable loads. To address load distribution complexity, Li et al. [27] developed an effective analytical method using the minimum energy principle, yielding loaded information consistent with practical applications. Blagojevic et al. [28] designed a compact double-stage cycloid reducer and analyzed multi-tooth contact stress. Lin et al. [29] proposed a double-stage cycloid mechanism via graph theory, conducting kinematic analysis under varied profile modifications. Hu et al. [30] presented an error sensitivity-based precision compensation method for RV reducers, validated through simulation and testing. Xu et al. [31] introduced dynamic measurement of lost motion precision and proposed a geometric lost motion separation approach.

In this context, this work proposes an integrated wear depth prediction model for RV reducers through synergistic integration of Archard's wear theory and a quasi-static mass-spring equivalent model. The framework quantitatively characterizes wear evolution in the cycloidal gear-pinwheel pair via TCA and LTCA by dynamically resolving contact forces, Hertzian stress, sliding distance, and pressure-dependent wear coefficients. This methodology integrates the Archard wear formulation with a 20-degree-of-freedom quasi-static model to simultaneously address progressive flank wear in the cycloidal-pin gear pair and multi-component error propagation. This approach overcomes a key limitation in existing literature, wherein wear and geometric errors are typically modeled as independent phenomena. Notably, the model incorporates an adaptive geometric recalibration mechanism triggered by wear thresholds and maps 11 manufacturing and assembly errors to equivalent spring deformations. Consequently, the model enables quantitative prediction of long-term precision degradation under coupled wear-geometric error interactions. Experimental-simulation co-validation further confirms that the framework resolves conflicts among tooth profile modification, contact stress amplification, and transmission accuracy loss, establishing a fundamental analytical tool for high-precision RV reducer design.

2 Quasi-static model for RV reducer considering wear depth

The methodology for establishing the quasi-static model of RV reducers, combined with the Archard wear model, is illustrated in Fig. 2. Input parameters for the quasi-static model and Archard wear model include geometric parameters of unworn cycloid gear surfaces, input transmission speed, operating load, material properties, errors of components, bearing clearances, and mesh stiffness of involute spur gears. Initially, the four parameters enclosed in the red box are inputted into the TCA model and Hertz contact model, yielding results such as contact points, backlash, curvature radius, Hertz contact stiffness, and Hertz contact half-width. These results are then inputted into the loaded distribution analysis to determine the mean mesh stiffness of cycloid-pin pairs and contact stress, with the mean mesh stiffness being utilized in the quasi-static model. Subsequently, the contact stress, combined with the relative sliding distance of cycloid-pin pairs obtained through kinematic analysis and wear coefficient calculated by an empirical formula, are inputted into the Archard wear model. The model assesses whether the current wear cycle meets the predetermined cycle number. If the conditions are met, the geometric parameters are updated and transmitted to the quasi-static model. If not, a further assessment is made to determine if the current accumulated wear depth surpasses the wear threshold. If it does, the geometric parameters are updated, and the process is repeated; otherwise, another wear cycle is initiated until the accumulated wear cycles meet the predetermined number. Then, the quasi-static system model is executed to perform relative analytical research. Finally, a dynamic transmission precision measurement based performance test platform of RV reducers is conducted to verify the correctness of the proposed model.

2.1 Wear depth model of cycloid-pin pairs based on LTCA

2.1.1 TCA and LTCA

According to the existing literature [32], TCA computes gear pair contact characteristics through mesh kinematics, necessitating compliance with:

1. Positional coincidence of contact point vectors between the driving and driven gears;
2. Their unit normal vectors are collinear (i.e. aligned in the same direction).

These conditions can be mathematically expressed as follows:

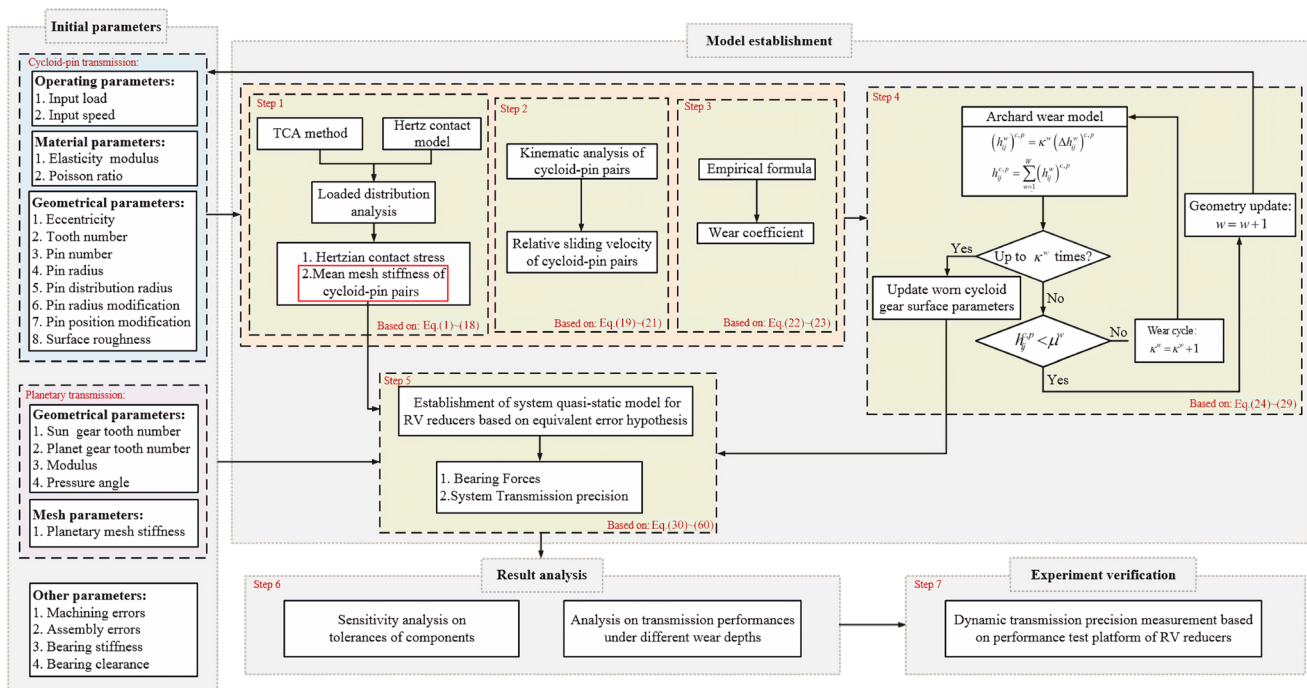


Fig. 2 Accuracy prediction model calculation flowchart

$$\begin{cases} \mathbf{r}_f^{(2)}(\phi_{ci}, \phi_{in}, \theta_{ci}) = \mathbf{r}_f^{(1)}(\theta_{pi}) \\ \mathbf{n}_f^{(2)}(\phi_{ci}, \theta_{ci}) = \mathbf{n}_f^{(1)}(\theta_{pi}) \end{cases} \quad (1)$$

where $\mathbf{r}_f^{(1)}$ and $\mathbf{r}_f^{(2)}$ represent the position vectors of the contact point on the cycloid gear and pins, respectively, defined in the fixed coordinate system of the pin gear. Similarly, $\mathbf{n}_f^{(1)}$ and $\mathbf{n}_f^{(2)}$ denote the unit normal vectors at the contact point. Given that the magnitude of the unit normal vectors is constrained to 1 ($\|\mathbf{n}_f^{(1)}\| = \|\mathbf{n}_f^{(2)}\| = 1$), the original vector equations reduce to three independent nonlinear equations. To solve this system, the input angle ϕ_{in} is prescribed as the known boundary condition, enabling the determination of the three remaining unknowns: the output angle of the cycloid gear (ϕ_{ci}), the angular position of the contact point on the cycloid gear (θ_{ci}), and the angular position of the contact point on the pins (θ_{pi}).

When external loads cause elastic deformation in cycloid-pin engagements, their interfacial behavior can be analyzed via Hertzian contact theory. For such line-contact problems, the half contact width a_{Hi} of the elastic contact zone on the i -th cycloid-pin pair, can be derived as follows [32]:

$$a_{Hi} = \sqrt{\frac{4F_i r^*}{\pi B E^*}} \quad (2)$$

where F_i is the load applied on the i -th cycloid-pin pair, B is the cycloid gear width, r^* is the relative radius of curvature of the cycloid-pin pair and E^* is the equivalent elastic modulus, where the detailed calculation methods can be found in literature [32].

Further, the magnitude of the contact deformation δ_i , the Hertzian contact stiffness k_i and Hertzian contact stress σ_{Hi} on the i -th cycloid-pin pair can be denoted as follows [32]:

$$\delta_i = \frac{2F_i}{\pi B} \left[\frac{1 - \nu_c^2}{E_c} \left(\ln \frac{4r_c}{a_{Hi}} - \frac{1}{2} \right) + \frac{1 - \nu_r^2}{E_r} \left(\ln \frac{4r_r}{a_{Hi}} - \frac{1}{2} \right) \right] \quad (3)$$

$$k_i = \frac{F_i}{\delta} = \frac{\pi B}{2 \left[\frac{1 - \nu_c^2}{E_c} \left(\ln \frac{4r_c}{a_{Hi}} - \frac{1}{2} \right) + \frac{1 - \nu_r^2}{E_r} \left(\ln \frac{4r_r}{a_{Hi}} - \frac{1}{2} \right) \right]} \quad (4)$$

$$\sigma_H = \frac{2F}{\pi a_{Hi} B} \quad (5)$$

where r_c and r_r correspond to the radius of curvature at the contact point for the cycloid gear and pin respectively; E_c and E_r represent the elastic modulus; ν_c and ν_r denote the poisson ratios.

In Fig. 3, the load distribution for each cycloid-pin pair is derivable by solving the equilibrium equations of all cycloid—pin tooth pairs, expressed as follows:

$$T_d - \sum (F_i l_i - F_{ui} l_{ui}) = 0 \quad (6)$$

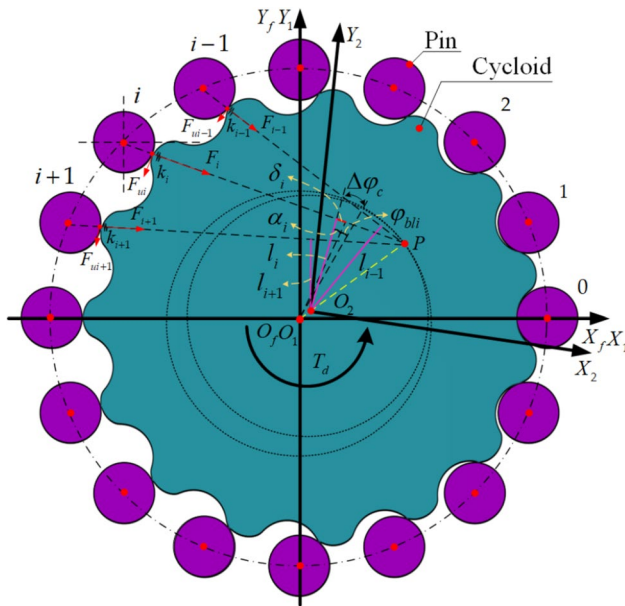


Fig. 3 Tooth contact model and load distribution analysis

where T_d is the external load; F_i is the force of the i -th cycloid-pin pair; F_{ui} is the friction force of the i -th cycloid-pin pair, defined as $F_{ui} = \mu_d F_i$, where μ_d is the sliding friction coefficient; l_i and l_{ui} are the moment arms of F_i and F_{ui} , respectively.

The force of the i -th cycloid-pin pair F_i can be expressed as follows:

$$F_i = k_i \delta_i = k_i l_i \alpha_i \quad (7)$$

where α_i is a micro angular displacement generated by δ_i , defined as follows:

$$\alpha_i = \begin{cases} \Delta\varphi_c - \phi_{bli}, & \Delta\varphi_c > \phi_{bli} \\ 0, & \Delta\varphi_c < \phi_{bli} \end{cases} \quad (8)$$

where $\Delta\varphi_c$ is the output angle variation and ϕ_{bli} is the backlash of the i -th cycloid-pin pair.

In this model, the tooth clearance ϕ_{bli} is not a fixed constant but a variable that dynamically changes with the input angle and pin index i . It represents the effective initial angular clearance generated along the line of action for the i -th gear pair at a specific moment. ϕ_{bli} is determined by the combined effects of the inherent design clearance resulting from tooth profile modifications and the equivalent displacement of geometric errors projected onto the line of action. These equivalent errors are quantified as initial displacements in the deformation compatibility analysis (Sect. 2.2.1). During each LTCA calculation, the instantaneous ϕ_{bli} is jointly

defined by the vector sum of all relevant equivalent error displacements and design modifications for the specific contact points determined by the TCA. Furthermore, as tooth surface wear progresses and geometric profiles iteratively update (Sect. 2.1.4), re-executed TCA/LTCA recalculates ϕ_{bli} under new contours, thereby incorporating wear effects on effective clearance. Consequently, ϕ_{bli} dynamically governs contact conditions and load distribution between pins, precisely capturing the complex interplay of error phase and wear evolution.

The moment arms l_i and l_{ui} can be derived as follows:

$$\begin{cases} l_i = \frac{a \sin(i\psi_r) + a \tan\theta_{pi} \cos(i\psi_r) - e \tan\theta_{pi} \cos\phi_{in} - e \sin\phi_{in}}{\sqrt{\tan^2\theta_{pi} + 1}} \\ l_{ui} = \frac{a \tan\theta_{pi} \sin(i\psi_r) - a \cos(i\psi_r) - e \tan\theta_{pi} \sin\phi_{in} + e \cos\phi_{in}}{\sqrt{\tan^2\theta_{pi} + 1}} - r_r \end{cases} \quad (9)$$

where a , ψ_r and e are the pin distribution radius, distribution angle of adjacent pins, eccentric distance.

Consequently, the Eq. (6) can be rewritten as follow:

$$T_d - \sum k_i \alpha_i l_i (l_i - \mu_d l_{ui}) = 0 \quad (10)$$

Numerical solution of the above equations in MATLAB yields the stress distribution, contact stress characteristics, transmission error profiles, and equivalent mesh forces between gear pairs. Additionally, interaction forces acting between turning-arm bearings and crankshafts can be analytically determined via subsystem decomposition methodology, wherein these components are divided into discrete subsystems for individual analysis.

As shown in Fig. 4(a), all cycloid-pin mesh forces converge at pitch point P , where frictional forces induce an additional torque on the cycloid gear. Simplification of the force system to pitch point P allows derivation of the resultant forces (ΣF_{ix} , ΣF_{iy}) and frictional torque component T_f through the following relationships:

$$\begin{cases} \Sigma F_{ix} = \Sigma F_i \frac{l_i}{ez_2} + \Sigma F_{ui} \left(\sqrt{1 - \left(\frac{l_i}{ez_2} \right)^2} \right) \\ \Sigma F_{iy} = \Sigma F_i \left(\sqrt{1 - \left(\frac{l_i}{ez_2} \right)^2} \right) + \Sigma F_{ui} \frac{l_i}{ez_2} \\ T_f = \Sigma F_{ui} l_{pi} \end{cases} \quad (11)$$

where z_2 denotes the tooth number of the cycloid gear, l_{pi} represents the distance between the contact point and pitch point P , defined as follows:

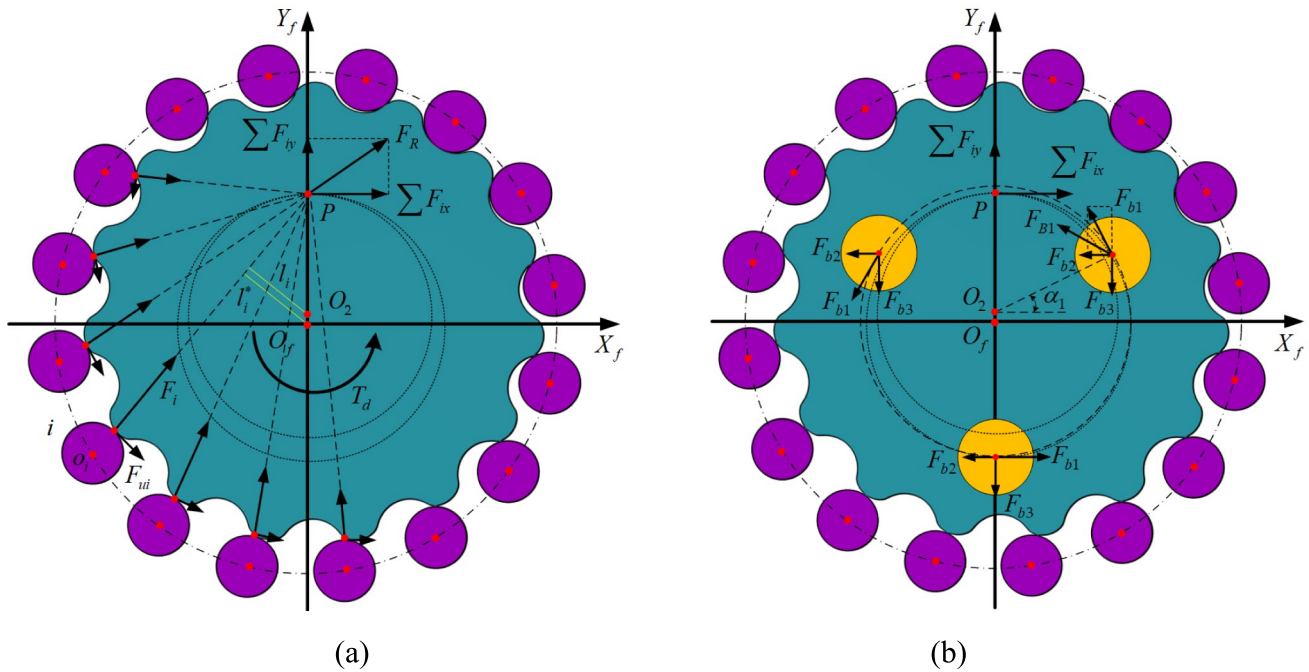


Fig. 4 Force analysis of needle gear cycloidal gear rotary arm bearing shaft: **a** Cycloid gear needle gear; **b** Cycloid wheel arm bearing

$$l_{pi} = \sqrt{(a \cos(i\psi_r) - ez_1 \cos \phi_{in})^2 + (a \sin(i\psi_r) - ez_1 \sin \phi_{in})^2} - r_r \quad (12)$$

where z_1 denotes the pin number.

In Fig. 4(b), following the simplification to geometric center O_2 , the resultant forces acting on each turning-arm bearing are resolved into three components: F_1 counteracts the torque generated by $\sum F_{ix}$ and T_f , F_2 balances $\sum F_{ix}$ and F_3 balances $\sum F_{iy}$, with their definitions expressed as follows:

$$\begin{cases} nF_{b1}r_{bg} - (ez_2 \sum F_{ix} - T_f) = 0 \\ nF_{b2} - \sum F_{ix} = 0 \\ nF_{b3} - \sum F_{iy} = 0 \end{cases} \quad (13)$$

where n is the number of the cranks.

Following calculation of the three constituent force components, vector synthesis is performed to resolve them into two principal orthogonal forces F_{BiX} and F_{BiY} along the axes x and y , ultimately yielding the resultant force F_{Bi} through the following vector composition relationships:

$$\begin{cases} F_{BiX} = F_{b2} + F_{b1} \sin \alpha_i = \frac{1}{n} \sum F_{ix} + \frac{ez_2}{r_{bg}} \sum F_{ix} \sin \alpha_i \\ F_{BiY} = F_{b3} - F_{b1} \cos \alpha_i = \frac{1}{n} \sum F_{iy} - \frac{ez_2}{r_{bg}} \sum F_{ix} \cos \alpha_i \end{cases}, \alpha_i = \frac{2\pi(i-1)}{n} + \phi_{in} \quad (14)$$

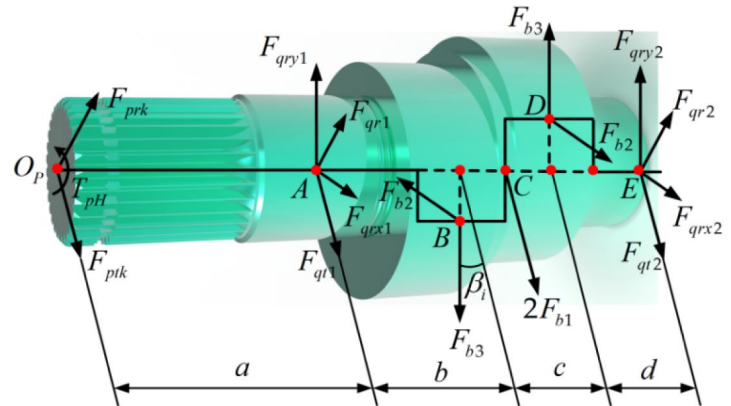
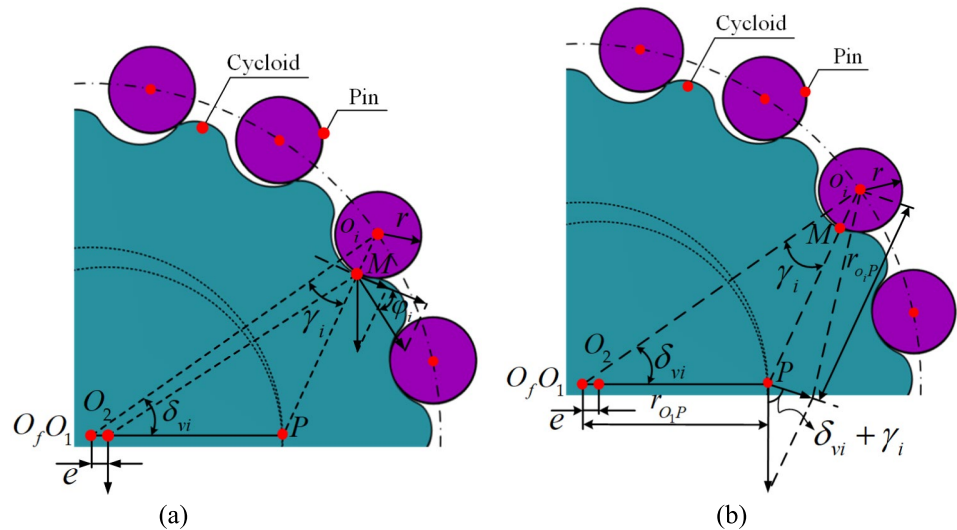
$$F_{Bi} = \sqrt{F_{BiX}^2 + F_{BiY}^2} \quad (15)$$

where r_{bg} is the distribution radius of the cranks on the cycloid gear.

Figure 5 schematically depicts the force distribution on individual cranks, where F_{prk} and F_{ptk} represent the radial and tangential force components from involute gear pairs. These cranks interface with planetary carriers at points A and E, exhibiting radial forces (F_{qr1} , F_{qr2}) and tangential forces (F_{qt1} , F_{qt2}). To simplify computation, the radial forces F_{qr1} and F_{qr2} are decomposed into orthogonal components along the eccentric direction (F_{qrx1} , F_{qrx2}) and perpendicular to it (F_{qry1} , F_{qry2}). By combining this analytical framework with static equilibrium conditions, all unknown forces are solved through the governing equations:

Fig. 5 Schematic diagram of force analysis of crankshaft

where r_{bg} is the distribution radius of the crankshafts on the cycloid gear.

**Fig. 6** Kinematic analysis: contact point on the **a** cycloid gear and **b** pin

$$\begin{cases}
 F_{ptk} = \frac{2F_{b2}e}{r_p} \\
 F_{prk} = F_{ptk} \tan \alpha_k \\
 F_{qt1} = -\frac{a+b+c+d}{b+c+d} F_{ptk} - \frac{2d+c}{b+c+d} F_{b1} \\
 F_{qt2} = \frac{a}{b+c+d} F_{ptk} - \frac{2b+c}{b+c+d} F_{b1} \\
 F_{qrx1} = \frac{c}{b+c+d} F_{b2} \\
 F_{qrx2} = -\frac{c}{b+c+d} F_{b2} \\
 F_{qry1} = -\frac{c}{b+c+d} F_{b3} \\
 F_{qry2} = -\frac{c}{b+c+d} F_{b3}
 \end{cases} \quad (16)$$

where r_p is the radius of reference circle of the sun gear, and α_k is the mesh angle of the involute pair.

Then, the resultant radial forces F_{qr1} and F_{qr2} can be represented as follows:

$$\begin{cases}
 F_{qr1} = \sqrt{F_{qrx1}^2 + F_{qry1}^2} \\
 F_{qr2} = \sqrt{F_{qrx2}^2 + F_{qry2}^2}
 \end{cases} \quad (17)$$

Consequently, the resultant forces F_{q1} and F_{q2} can be represented as follows:

$$\begin{cases}
 F_{q1} = \sqrt{F_{qr1}^2 + F_{qt1}^2} \\
 F_{q2} = \sqrt{F_{qr2}^2 + F_{qt2}^2}
 \end{cases} \quad (18)$$

2.1.2 Relative sliding distance

Before computing cycloid-pin pairs wear depth, kinematic analysis of their contact point is essential to determine rolling and sliding velocities. In Fig. 6, symbol M means the common contact point. The velocity of M on the cycloid gear is recorded as V_c , and the velocity on the pin is recorded as V_p . Both V_c and V_p are at the common tangent line of the contact point M . V_c denotes the projection of the resultant of

its rotation velocity around the pitch point P and its translational velocity. V_p is the rotation velocity of the pitch point P around the center o_i of the i -th pin. These velocities obey the following equations:

$$\begin{cases} V_c = \omega_{out} \cos \varphi_i \rho_{ci} - \omega_{in} \cos (\delta_{vi} + \gamma_i) e + \omega_{in} \cos (\delta_{vi} + \gamma_i) \frac{r_{O_1 P} \rho_{PP}}{r_{O_1 P}} \\ V_p = \omega_{in} \cos (\delta_{vi} + \gamma_i) \frac{r_{O_1 P} \rho_{PP}}{r_{O_1 P}} \end{cases} \quad (19)$$

Thus, the relative velocity V_r is expressed as follows:

$$V_r = V_c - V_p = \omega_{out} \cos \varphi_i \rho_{ci} - \omega_{in} \cos (\delta_{vi} + \gamma_i) e \quad (20)$$

where ω_{in} and ω_{out} represent the input angular velocity and output angular velocity, respectively. The output angular velocity can be derived from $\omega_{out} = \omega_{in} / z_2$. δ_{vi} represents the angle between $O_1 o_i$ and $O_1 P$, γ_i represents the angle between $O_1 o_i$ and $o_i P$, φ_i is the angle between the velocity of point M and the tangent velocity.

Then, based on the single point observation method [2], the sliding distances s_c and s_p of the cycloid gear and pin are presented as follows:

$$s_c = 2a_{Hi} \left| \frac{V_r}{V_c} \right|, s_p = 2a_{Hi} \left| \frac{V_r}{V_p} \right| \quad (21)$$

2.1.3 Wear coefficient

The wear coefficient k is affected by some parameters, like operating speed, applied load, surface roughness, lubrication state and so on, and obtained by an empirical formula [33] as follows:

$$k = \frac{3.981 \times 10^{29}}{E^*} L^{1.219} G^{-7.377} S^{1.589} \quad (22)$$

where E^* denotes the equivalent elasticity modulus. L , G and S stand for the dimensionless load, dimensionless lubrication pressure viscosity coefficient, and dimensionless composite roughness, respectively. These are calculated as follows:

$$L = \frac{F}{E^* r^*}, G = \alpha E^*, S = \frac{\sqrt{R_{\alpha 1}^2 + R_{\alpha 2}^2}}{r^*} \quad (23)$$

where F stands for the load applied in the unit length, α is the pressure viscosity coefficient, which is temperature-dependent. r^* signifies the relative radius of curvature of the cycloid-pin pair. $R_{\alpha 1}$ and $R_{\alpha 2}$ are the surface roughness values of the cycloid gear and the pin, respectively.

2.1.4 Wear depth model

While surface discretization facilitates precise stress mapping, the core innovation lies in its integration within a predictive, dynamically updated framework. Unlike static models, the mesh nodes in our approach serve as carriers for time-evolving wear depth data. Crucially, this discretization scheme is coupled with a wear threshold mechanism, enabling iterative recalibration of the tooth surface geometry within the TCA/LTCA loop.

Furthermore, discretization along the tooth width direction is implemented to preserve model generality and structural completeness. Although the current study assumes ideal alignment and does not explicitly incorporate axial modifications or errors, the two-dimensional grid formulation inherently accommodates potential non-uniform contact pressure distributions across the tooth face. This approach provides a rigorous foundation that can seamlessly integrate future investigations into axial misalignments, edge-loading effects, or deliberate axial profile modifications, moving beyond the limitations of a simplified line-contact model.

Archard's wear equation, as the widespread used wear model [1], is given as follows:

$$\frac{V}{S} = K \frac{P_r}{W} \quad (24)$$

where V represents the wear volume of the material, S is the relative sliding distance, P_r denotes the contact normal pressure, W is the material hardness, and K is the dimensionless wear coefficient.

Assuming that the material hardness of the cycloidal drill cylinder remains constant during the wear process, the single-point wear calculation formula on the contact tooth surface can be expressed as:

$$\frac{dh}{ds} = kp \quad (25)$$

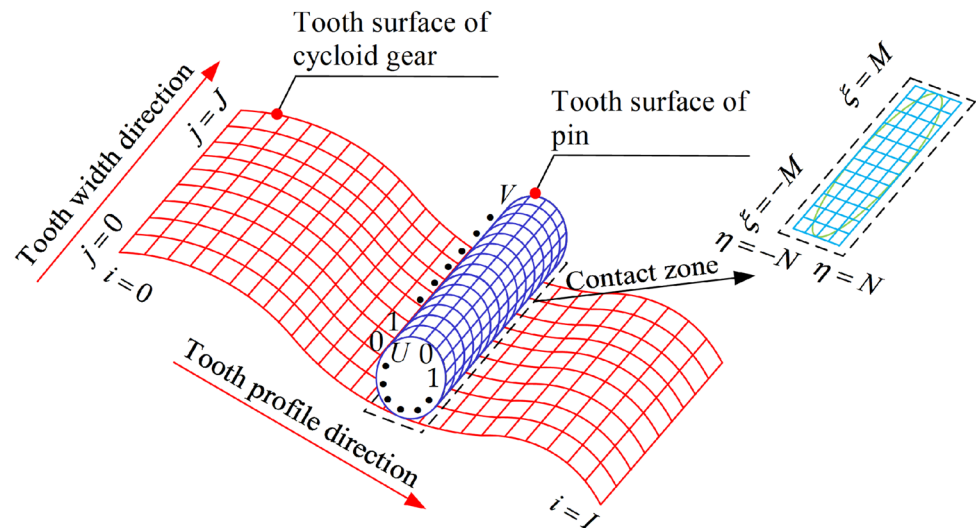
where h is the wear depth. k , p and s represent the wear coefficient, contact stress and sliding distance of the cycloid-pin pairs respectively, described in the previous part.

Furthermore, the above equation can be expressed in integral form as:

$$h = \int_0^s kp ds \quad (26)$$

To enhance precision, the cycloid gear tooth surfaces are discretized into coarse grids with dimensions $I \times J$ along the tooth profile and gear width. Similarly, pins are discretized into corresponding coarse grids with dimensions $U \times V$

Fig. 7 Tooth surface grid segmentation for the cycloid-pin pair



along the pin profile and pin width. Subsequently, contact areas are further refined into fine grids, denoted by the dimensions $2N \times 2M$ along the tooth profile and gear width, as illustrated in Fig. 7. Coarse grids remain stationary on the cycloid gears and pins, while fine grids move with the rotation of the cycloid-pin pair to identify every contact zone. The contact point is easily determined by matching coarse grid nodes with fine grid nodes, enabling precise stress distribution analysis. By sufficiently reducing the size of fine grids and ensuring overlap with coarse grids, the contact stress at a coarse grid node ij can be characterized by the mean contact stress at a corresponding fine grid node $\eta\zeta$. This approach facilitates the calculation of mean stress at the coarse grid node ij during a mesh cycle.

A wear threshold μ^w is set to improve the computation efficiency. When wear depth exceeds the threshold μ^w , tooth geometry parameters must be updated within the TCA model to compute revised stress distributions; otherwise, the distribution remains unchanged. After achieving the stress p , sliding distance s and wear coefficient k , local wear depth at any tooth position is determined through numerical integration of Eq. (26):

$$(\Delta h_{ij}^w)^{c,p} = k (\overline{p_{ij}^w})^{c,p} (s_{ij}^w)^{c,p} \quad (27)$$

where the outer layer superscript c and p denote the cycloid gear and pin, the inner layer superscript w denotes the ordinal number for updating the geometrical parameters, the subscript ij stand for the coarse grid node in contact, Δh denotes the wear depth and p is the mean stress.

Suppose continuously applying the Eq. (27) for κ^w times, it be written as follows:

$$(h_{ij}^w)^{c,p} = \kappa^w (\Delta h_{ij}^w)^{c,p} \quad (28)$$

Then repeat the above steps for the number of w ($w=0, 1, 2, \dots, W$) times, the accumulated wear depth can be presented as:

$$h_{ij}^{c,p} = \sum_{w=1}^W (h_{ij}^w)^{c,p} \quad (29)$$

2.2 Establishment of quasi-static model for RV reducer

The proposed quasi-static model is designed to explicitly account for the influence of geometric errors and progressive tooth surface wear on the system's static equilibrium and transmission accuracy. Geometric errors are mapped into equivalent initial deformations or displacements along the lines of action of the corresponding spring elements via the mass-spring equivalence method. Tooth surface wear, calculated through the integrated Archard-LTCA framework, dynamically alters the geometry of the cycloid-pin pair. When accumulated wear at a mesh node exceeds a pre-defined threshold, the tooth profile parameters are updated. This triggers a recalculation of contact conditions, contact stiffness, and load distribution within the LTCA. The updated geometric parameters and stiffness values are then fed back into the quasi-static model, influencing the deformation compatibility equations and force equilibrium. Crucially, the Hertzian contact stiffness k_i for each cycloid-pin pair is recalculated within LTCA based on the instantaneous contact geometry, thereby indirectly incorporating the effect of wear on mesh stiffness through the "geometry update—contact point change—stiffness recalculation" chain. This integrated approach enables the model to simulate long-term transmission accuracy degradation under coupled error-wear interactions.

To simplify the quasi-static model for the complex structure an RV reducer, certain steps are taken, and the following preconditions are emphasized for simplicity:

- (1) The rotation angular of the sun gear is constant;
- (2) The mesh stiffness and supporting bearing stiffness are equivalent as stiffness of a linear spring;
- (3) Neglect the effect of the mass acceleration of all the parts in the model;
- (4) Neglect the damping and friction on the contact surface;
- (5) Only errors, deformations and micro-displacements in the planar surface are considered;
- (6) The stiffness component of all the bearings along the axis and are equal.

The quasi-static mathematical model with an explosion diagram is depicted in Fig. 8, with symbols s , p , h , c and o representing the sun gear, planetary gears, cranks, cycloid gears, and output disc.

The coordinate system $S(xoy)$ should be fastened to the pinwheel, and the movable coordinate system $S_k (k=s, p, h, c, o; i=1,2,3; j=1,2)$ should be set at the component center.

Symbols i and j denote the ordinal number of planetary and cycloid gears.

The support stiffness of the sun gear and output disc is denoted as k_s and k_o . k_{bcij} represents the support stiffness between i -th crank and the j -th cycloid gear, while k_{boi} is the support stiffness between the i -th crank and output disc. Symbols k_{spi} and k_{cpjg} denote mesh stiffness of the sun gear and i -th planetary gear, and the j -th cycloid gear and pin. Symbol θ_s is the sun gear rotation angle. θ_p and θ_{pi} denote the theoretical and actual rotation angles of the planetary gears while θ_o and θ_{or} corresponding to those of the output disc. θ_a and θ_{aj} represent the actual revolution angle and actual rotation angle of the j -th cycloid gear. Symbols X_s , Y_s , X_{pi} , Y_{pi} , η_j , X_o and Y_o mean linear micro-displacements of the sun gear, i -th planetary gear, j -th cycloid gear and output disc, while $\delta\theta_{pi}(\delta\theta_{pi}=\theta_{pi}-\theta_p)$, $\delta\theta_{dj}(\delta\theta_{dj}=\theta_j-\theta_o)$, $\delta\theta_{doj}(\delta\theta_{doj}=\theta_{aj}-\theta_p)$ and $\delta\theta_{or}(\delta\theta_{or}=\theta_{or}-\theta_o)$ denote the angular micro-displacement of the i -th planetary gear, angular micro-displacement for rotation and revolution of j -th cycloid gear and output disc. Phase angles of the planetary

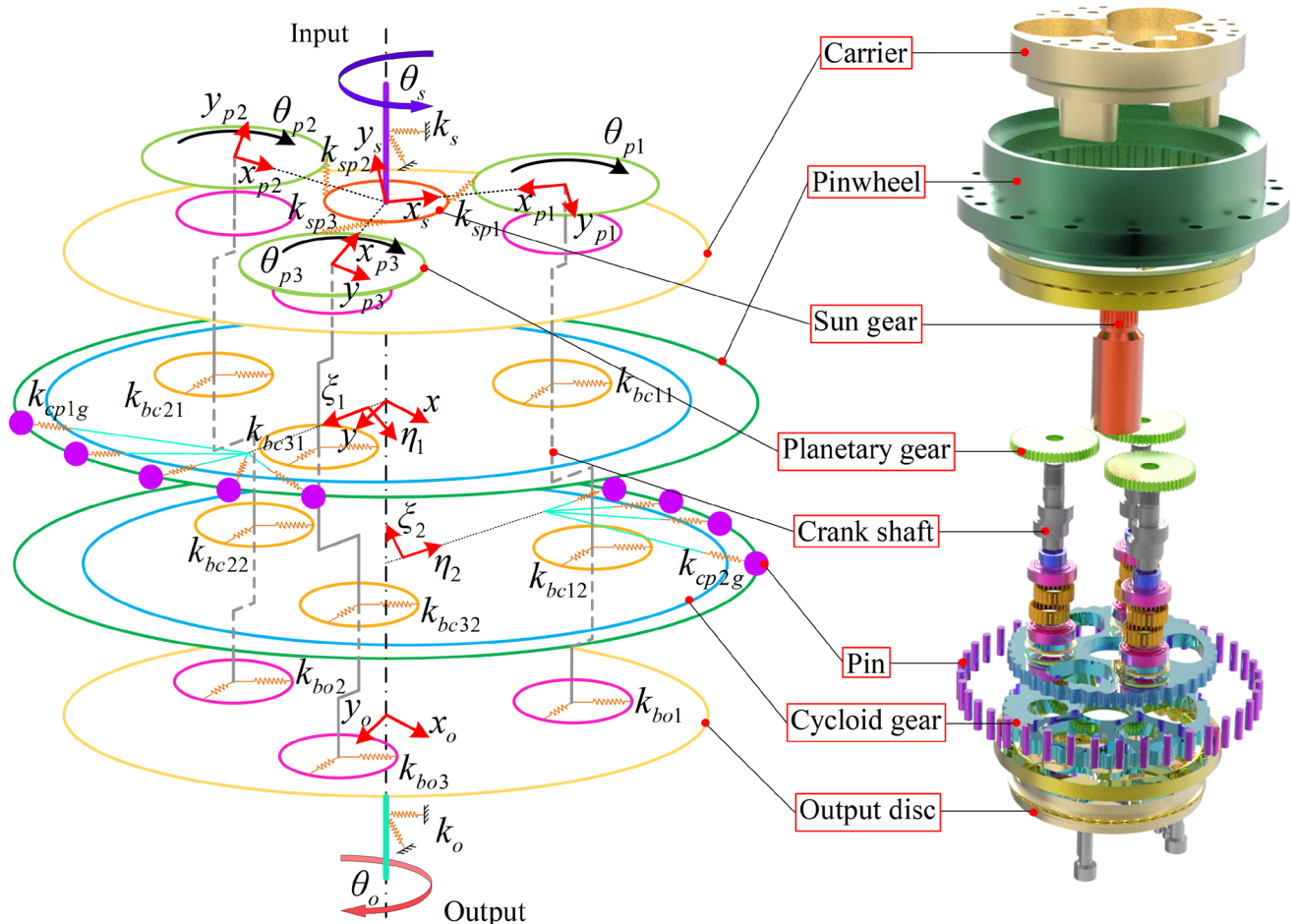


Fig. 8 Quasi-static model diagram for RV reducers

and cycloid gears can be recorded as φ_i and ψ_j . Hence, the quasi-static model has 20 degrees of freedom, as follows:

$$F = \{X_s, Y_s, X_{p1}, Y_{p1}, \delta\theta_{p1}, X_{p2}, Y_{p2}, \delta\theta_{p2}, X_{p3}, Y_{p3}, \delta\theta_{p3}, \eta_1, \delta\theta_{d1}, \delta\theta_{do1}, \eta_2, \delta\theta_{d2}, \delta\theta_{do2}, X_o, Y_o, \delta\theta_{or}\} \quad (30)$$

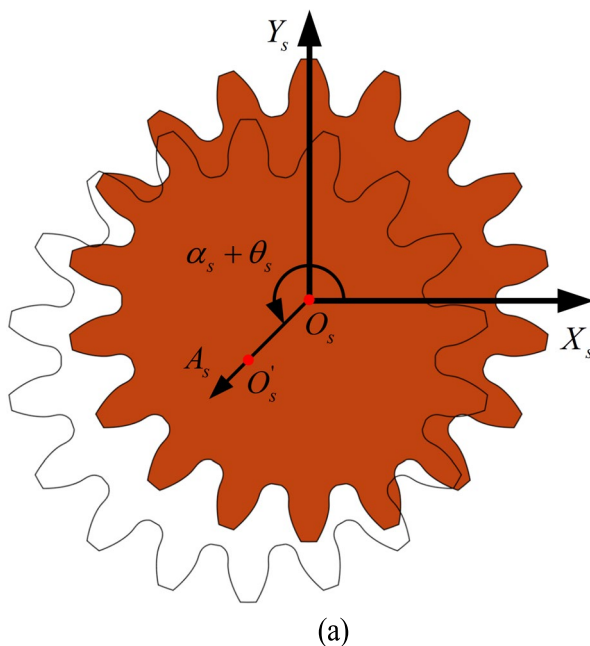
2.2.1 Deformation compatibility analysis

Before deriving the mechanical equilibrium equations, deformation-compatibility analysis should be conducted to connect the deformations of parts with errors to the load applied on the parts.

1) Analysis on the equivalent supporting spring stiffness of the sun gear at the line of action.

Mounting the sun gear on the motor output shaft introduces positional deviations, which propagate to assembly errors during gear installation. Consequently, deformations in the sun gear's support spring are resolved into the axes x and y , recorded as $(\delta X_s, \delta Y_s)$. Figure 9(a) illustrates the sun gear assembly error (A_s, α_s) , where A_s denotes the absolute error magnitude and α_s represents the error phase angle relative to the theoretical position. Thus, the equivalent error $(\sigma_{sx}, \sigma_{sy})$ resulted from (A_s, α_s) is decomposed into the axes x and y as follows:

$$\begin{cases} \sigma_{sx} = A_s \cos(\alpha_s + \theta_s) \\ \sigma_{sy} = A_s \sin(\alpha_s + \theta_s) \end{cases} \quad (31)$$



Then, the deformation-compatibility equations should be written as below:

$$\begin{cases} \delta X_s = X_s - \sigma_{sx} \\ \delta Y_s = Y_s - \sigma_{sy} \end{cases} \quad (32)$$

2) Analysis on the sun gear and planetary gear at the engagement line.

Equivalent deformation at the equivalent spring engagement line between the sun gear and i -th planetary gear is quantified as δsp_i . Figure 9(b) illustrates the machining errors for these components, characterized by (E_s, β_s) for the sun gear and (E_{pi}, β_{pi}) for the planetary gear. e_s and e_{pi} are the equivalent errors at the mesh line generated by the machining errors, written as follows:

$$\begin{cases} e_{si} = -E_s \cos(\theta_s + \beta_s - A_i) \\ e_{pi} = E_{pi} \cos(-\theta_{pi} + \beta_{pi} - A_i) \end{cases} \quad (33)$$

where $A_i = \pi/2 - \alpha + \theta_c + \varphi_i$ and α is the mesh angle for the involute spur gears.

The deformation δs_i of the i -th gear pair led by the sun gear is written as below:

$$\delta s_i = X_s \cos A_i + Y_s \sin A_i \quad (34)$$

The deformation δp_i led by the planetary gear can be written as follows:

$$\delta p_i = -r_{db}(\theta_{pi} - \theta_p) - X_{pi} \cos A_i - Y_{pi} \sin A_i \quad (35)$$

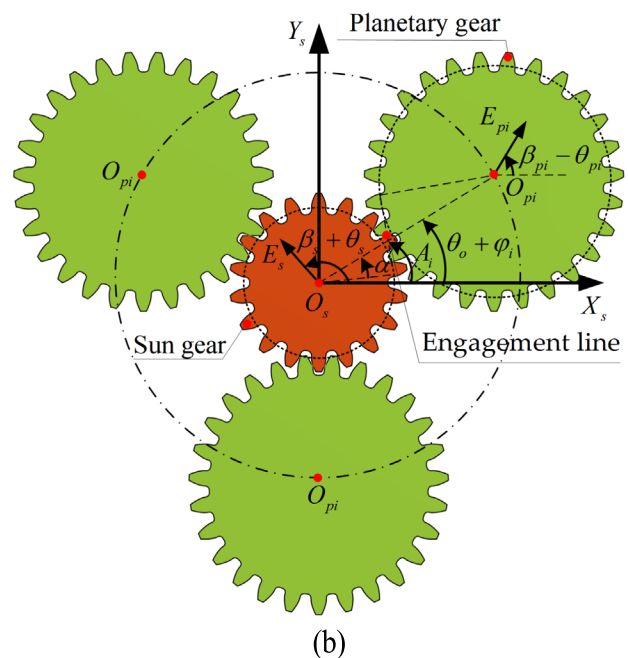


Fig. 9 Equivalent errors between the sun and planetary gear: **a** assembly error of the sun gear and **b** error along the engagement line

where r_{db} is the base circle radius.

Then, the deformation-compatibility equations can be written as follows:

$$\delta sp_i = \delta s_i + \delta p_i - e_{si} - e_{pi} \quad (36)$$

3) Analysis on the crank and cycloid gear at the line of action.

Equivalent spring deformations between the i -th crank and j -th cycloid gear are resolved into the axes x and y , recorded as $(\delta X_{bcij}, \delta Y_{bcij})$. Figure 10 characterizes positional deviations at the i -th hole of the j -th cycloid gear, expressed as (E_{hij}, β_{hij}) , which generates the equivalent error as follows:

$$\begin{cases} e_{hxi} = E_{hij} \cos(\theta_c + \varphi_i + \beta_{hij}) \\ e_{hyi} = E_{hij} \sin(\theta_c + \varphi_i + \beta_{hij}) \end{cases} \quad (37)$$

Similarly, the eccentric error of the j -th cam on the i -th crank is expressed with (E_{cij}, β_{cij}) leading to the equivalent error as follows:

$$\begin{cases} e_{cxi} = E_{cij} \cos(\theta_p + \varphi_j + \beta_{cij}) \\ e_{cyi} = -E_{cij} \sin(\theta_p + \varphi_j + \beta_{cij}) \end{cases} \quad (38)$$

And the equivalent deformation led by the three degrees of freedom of cycloid gears decomposed into the axes x and y are written as follows:

$$\begin{cases} \delta X_{cij} = e\delta\theta_{doj} \sin(\theta_p + \varphi_j) + R_c\delta\theta_j \sin(\theta_c + \varphi_j) - \eta_j \cos(\theta_p + \varphi_j) \\ \delta Y_{cij} = e\delta\theta_{doj} \cos(\theta_p + \varphi_j) - R_c\delta\theta_j \cos(\theta_c + \varphi_j) + \eta_j \sin(\theta_p + \varphi_j) \end{cases} \quad (39)$$

where R_c is the central distance of the planetary transmission, and e is the eccentric distance.

Likewise, the equivalent deformation generated by the three degrees of freedom of crankshafts decomposed into the axes x and y are written as follows:

$$\begin{cases} \delta X_{qij} = -e\delta\theta_{pi} \sin(\theta_p + \varphi_j) + X_{pi} \\ \delta Y_{qij} = -e\delta\theta_{pi} \cos(\theta_p + \varphi_j) + Y_{pi} \end{cases} \quad (40)$$

Thus, the deformation-compatibility equations should be presented as below:

$$\begin{cases} \delta X_{bcij} = \delta X_{cij} + \delta X_{qij} + e_{cxi} - e_{hxi} \\ \delta Y_{bcij} = \delta Y_{cij} + \delta Y_{qij} + e_{cyi} - e_{hyi} \end{cases} \quad (41)$$

4) Analysis on the j -th cycloid gear and g -th pin at the engagement line.

The equivalent deformation between the j -th cycloid gear and g -th pin is recorded as δd_{cpjg} at the direction of the engagement line. In Fig. 11, the error generated by the pin radius modification and pin distribution radius modification is written as follows:

$$\begin{cases} \delta E_j = \Delta\rho_p + \Delta a \cos \tau \\ \cos \tau = \frac{1 - \lambda_m \cos \phi_{Pjg}}{\sqrt{1 + \lambda_m^2 - 2\lambda_m \cos \phi_{Pjg}}} \end{cases} \quad (42)$$

where λ_m is the short amplitude coefficient, Δa means the amount of pin distribution radius modification and $\Delta\rho_p$ is the amount of pin radius modification.

The equivalent errors e_{Rjg} and e_{ARjg} generated by the radial tooth profile deviation e_{Rjg} on the cycloid gear and

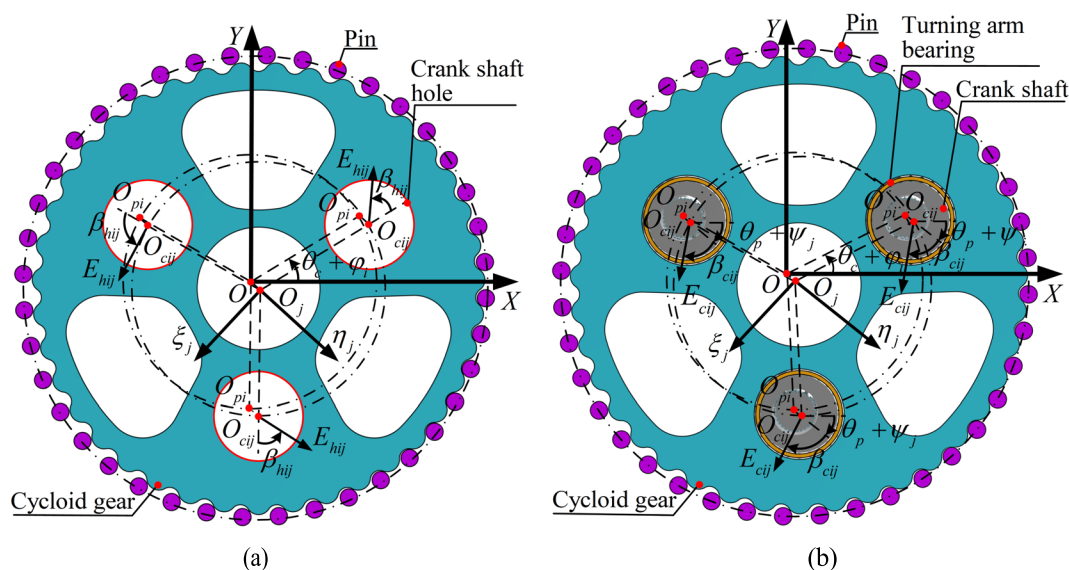


Fig. 10 Equivalent errors between the cycloid gear and crankshaft: **a** position error of crankshaft holes and **b** cam eccentric error of crankshafts

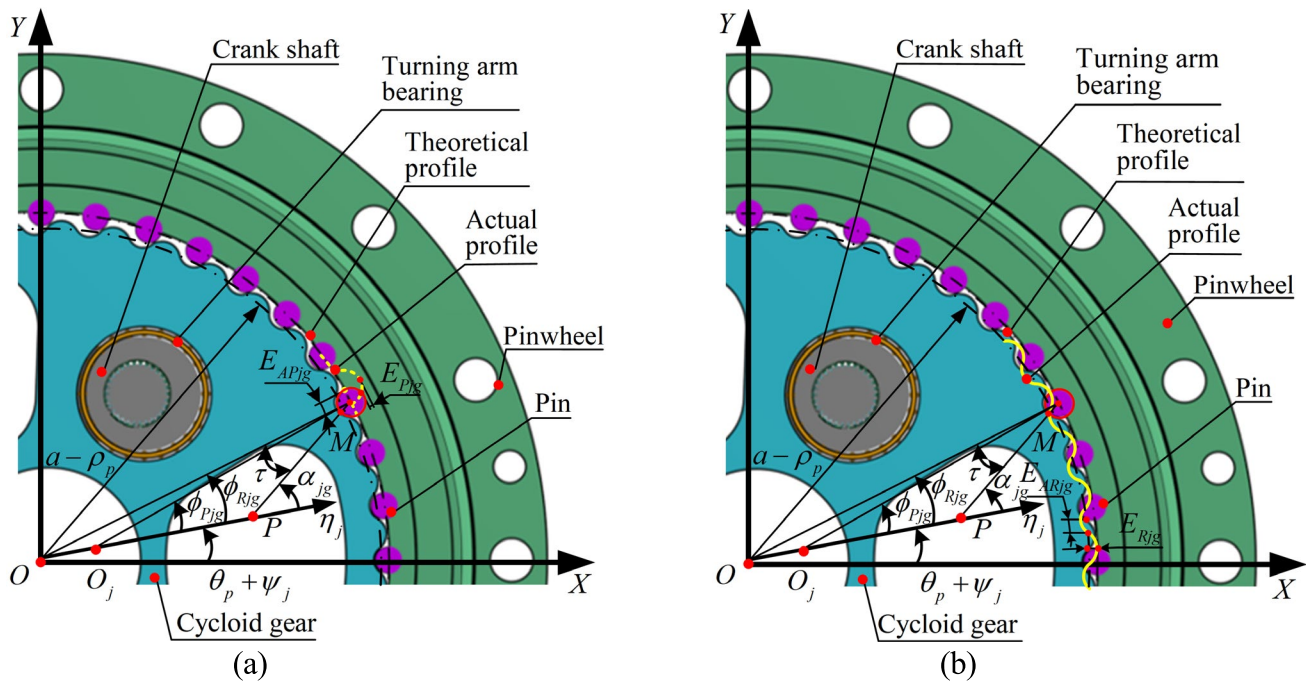


Fig. 11 Equivalent errors between the cycloid gear and pinwheel: **a** radial tooth profile deviation and accumulated pitch deviation on the cycloid gear and **b** radial tooth profile deviation and accumulated pitch deviation of the pin groove on the pinwheel

accumulated tooth pitch deviation e_{ARjg} on the cycloid gear at the engagement line can be presented as below:

$$\begin{cases} e_{Rjg} = -E_{Rjg} \cos(\alpha_{jg} - \phi_{Pjg}) \\ e_{ARjg} = -E_{ARjg} \sin(\alpha_{jg} - \phi_{Pjg}) \end{cases} \quad (43)$$

where α_{jg} denotes the angle between the line from the pitch point of the j -th cycloid gear to the g -th pin center and the reference axis η_j . ϕ_{Pjg} represents the angle connecting the line from the j -th cycloid gear center to the g -th pin center and the same axis.

The equivalent errors e_{Pjg} and e_{APjg} generated by the radial tooth profile deviation e_{Pjg} and accumulated tooth pitch deviation e_{APjg} at the mesh line are as follows:

$$\begin{cases} e_{Pjg} = E_{Pjg} \cos(\alpha_{jg} - \phi_{Rjg}) \\ e_{APjg} = E_{APjg} \sin(\alpha_{jg} - \phi_{Rjg}) \end{cases} \quad (44)$$

where ϕ_{Rjg} is the angle of the radial direction of the g -th pin and axis η_j .

Then, the equivalent deformation led by the three degrees of freedom of cycloid gears at the engagement line is written as follows:

$$\delta c_{jg} = e\delta\theta_{doj} \sin \alpha_{jg} - R_d\delta\theta_{dj} \sin \alpha_{jg} + \eta_j \cos \alpha_{jg} \quad (45)$$

Thus, the deformation-compatibility equations can be written as follows:

$$\delta d_{cpjg} = \delta c_{jg} - \delta E_j - e_{Rjg} - e_{ARjg} - e_{Pjg} - e_{APjg} \quad (46)$$

5) Analysis on the crank and output disc at the line of action.

Equivalent deformations of the i -th crank and output disc are resolved into the axes x and y , recorded as (X_{boi}, β_{boi}) . Figure 12a depicts the positional deviation at the i -th bearing hole, represented as (X_{boi}, β_{boi}) , which generates the equivalent errors e_{boxi} and e_{boyi} as follows:

$$\begin{cases} e_{boxi} = E_{boi} \cos(\theta_c + \varphi_i + \beta_{boi}) \\ e_{boyi} = E_{boi} \sin(\theta_c + \varphi_i + \beta_{boi}) \end{cases} \quad (47)$$

Then, the deformation-compatibility equations between the i -th crank and output disc at the line of action can be written as follows:

$$\begin{cases} \delta X_{boi} = R_c\delta\theta_{or} \sin(\theta_o + \phi_i) + X_{pi} - X_{or} - e_{boxi} \\ \delta Y_{boi} = -R_c\delta\theta_{or} \cos(\theta_o + \phi_i) + Y_{pi} - Y_{or} - e_{boyi} \end{cases} \quad (48)$$

6) Analysis on the output disc and pinwheel at the line of action.

Deformations of the equivalent spring connecting the output disc and pinwheel are resolved into the axes x and y , denoted as $(\delta X_o, \delta \beta_o)$. Figure 12b illustrates the output disc assembly error (A_o, α_o) , which generates the equivalent errors σ_{ox} and σ_{oy} as follows:

$$\begin{cases} \sigma_{ox} = A_o \cos(\alpha_o + \theta_o) \\ \sigma_{oy} = A_o \sin(\alpha_o + \theta_o) \end{cases} \quad (49)$$

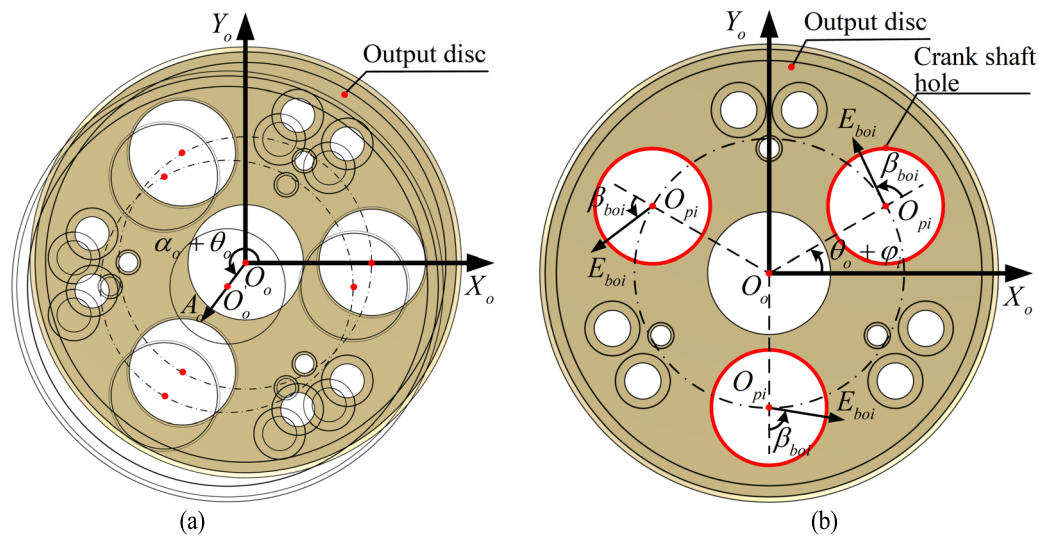


Fig. 12 Equivalent errors on the output disc: **a** assembly error and **b** position error of crankshaft

Thus, the deformation-compatibility equations can be written as follows:

$$\begin{cases} \sigma X_o = X_o - \sigma_{ox} \\ \sigma Y_o = Y_o - \sigma_{oy} \end{cases} \quad (50)$$

2.2.2 Force equilibrium analysis and establishment of force equilibrium equations

(1) Force analysis on the sun gear

The equivalent spring forces along the axes x and y at the supporting bearings and the mesh force are as follows:

$$\begin{cases} F_{sx} = k_s \delta X_s \\ F_{sy} = k_s \delta Y_s \end{cases}, F_{spi} = k_{spi} \delta_{spi} \quad (51)$$

Thus, the equilibrium equations of the sun gear are given as follow:

$$\begin{cases} \sum_{i=1}^n F_{spi} \cos A_i + F_{sx} = 0 \\ \sum_{i=1}^n F_{spi} \sin A_i + F_{sy} = 0 \end{cases} \quad (52)$$

2) Force analysis on the cranks

The equivalent forces of the i -th crank and j -th cycloid gear along the axes x and y are as follows:

$$\begin{cases} F_{bcxi} = k_{bcij} \delta X_{bcij} \\ F_{bcyi} = k_{bcij} \delta Y_{bcij} \end{cases} \quad (53)$$

The equivalent forces between the i -th crank and output disc along the axes x and y are as follows:

$$\begin{cases} F_{boxi} = k_{boi} \delta X_{boi} \\ F_{boyi} = k_{boi} \delta Y_{boi} \end{cases} \quad (54)$$

Then, the equilibrium equations of the crank are as follow:

$$\begin{cases} F_{spi} r_{gp} + e \sum_{j=1}^2 [F_{bcxij} \sin(\theta_p + \phi_j) + F_{bcyij} \cos(\theta_p + \phi_j)] = 0 \\ F_{spi} \cos A_i - \sum_{j=1}^2 F_{bcxij} - F_{boxi} = 0 \\ F_{spi} \sin A_i - \sum_{j=1}^2 F_{bcyij} - F_{boyi} = 0 \end{cases} \quad (55)$$

3) Force analysis on the cycloid gears

The mesh force between the i -th cycloid gear and g -th pin is as follows:

$$F_{cpjg} = k_{cpjg} \delta d_{cpjg} \quad (56)$$

where, when $\delta d_{cpjg} < 0$, the cycloid-pin pair is considered out of contact and then $k_{cpjg} = 0$.

Then, the equilibrium equations of the j -th cycloid gear are given as follow:

$$\begin{cases} \sum_{i=1}^n [F_{bcyij} \sin(\theta_p + \psi_j) - F_{bcxij} \cos(\theta_p + \psi_j)] + \sum_{g=1}^{z_n} (F_{cpjg} \cos \alpha_{jg}) = 0 \\ R_c \sum_{i=1}^n [F_{bcyij} \cos(\theta_c + \varphi_i) - F_{bcxij} \sin(\theta_c + \varphi_i)] + R_d \sum_{g=1}^{z_n} (F_{cpjg} \sin \alpha_{jg}) = 0 \\ e \sum_{i=1}^n [F_{bcyij} \cos(\theta_p + \psi_j) + F_{bcxij} \sin(\theta_p + \psi_j)] + e \sum_{g=1}^{z_n} (F_{cpjg} \sin \alpha_{jg}) = 0 \end{cases} \quad (57)$$

where, z_n is the number of the pins and R_d is the pitch distance of the cycloid gear.

4) Force analysis on the output disc

The forces between the i -th crank and output disc along the axes x and y are as follows:

$$\begin{cases} F_{boxi} = k_{boi} \delta X_{boi} \\ F_{boyi} = k_{boi} \delta Y_{boi} \end{cases} \quad (58)$$

Then, the equilibrium equations of the output disc are given as follow:

$$\begin{cases} R_c \sum_{i=1}^n [F_{boxi} \sin(\theta_c + \phi_i) - F_{boyi} \cos(\theta_c + \phi_i)] = -T_{out} \\ \sum_{i=1}^n F_{boxi} - k_o \delta X_o = 0 \\ \sum_{i=1}^n F_{boyi} - k_o \delta Y_o = 0 \end{cases} \quad (59)$$

where, the output disc bears a torque of T_{out} .

Hence, the system equilibrium equations are as follows:

Table 1 Primary parameters of the RV reducer

	Parameter	Value
Primary involute gear transmission stage	Module	1.5
	Pressure angle (°)	20
	Sun gear teeth	17
	Planet gear teeth	61
	Gear width (mm)	12
	Elastic modulus (GPa)	206
	Poisson's ratio	0.3
Secondary cycloid-pin gear transmission stage	Pin number	40
	Cycloid gear teeth	39
	Pin radius (mm)	4
	Pin distribution radius (mm)	111.5
	Eccentric distance (mm)	2
	Equidistant modification (μm)	8
	Profile shift modification (μm)	10
	Cycloid gear width (mm)	20
	Elastic modulus (GPa)	206
	Poisson's ratio	0.3
	Cycloid gear surface roughness (μm)	0.05
	Pin surface roughness (μm)	0.05
	Pressure viscosity coefficient (MPa-1)	0.0198
	Load torque (Nm)	5000
Total transmission ratio		144.53

$$\begin{cases} \sum_{i=1}^n F_{spi} \cos A_i + F_{sx} = 0 \\ \sum_{i=1}^n F_{spi} \sin A_i + F_{sy} = 0 \\ F_{spi} r_{gp} + e \sum_{j=1}^2 [F_{bcxij} \sin(\theta_p + \psi_j) + F_{bcyij} \cos(\theta_p + \psi_j)] = 0 \\ F_{spi} \cos A_i - \sum_{j=1}^2 F_{bcxij} - F_{boxi} = 0 \\ F_{spi} \sin A_i - \sum_{j=1}^2 F_{bcyij} - F_{boyi} = 0 \\ \sum_{i=1}^n [F_{bcyij} \sin(\theta_p + \psi_j) - F_{bcxij} \cos(\theta_p + \psi_j)] + \sum_{g=1}^{z_n} (F_{cpjg} \cos \alpha_{jg}) = 0 \\ R_c \sum_{i=1}^n [F_{bcyij} \cos(\theta_c + \varphi_i) - F_{bcxij} \sin(\theta_c + \varphi_i)] + R_d \sum_{g=1}^{z_n} F_{cpjg} \sin \alpha_{jg} = 0 \\ e \sum_{i=1}^n [F_{bcyij} \cos(\theta_p + \psi_j) + F_{bcxij} \sin(\theta_p + \psi_j)] + e \sum_{g=1}^{z_n} F_{cpjg} \sin \alpha_{jg} = 0 \\ R_c \sum_{i=1}^n [F_{boxi} \sin(\theta_c + \varphi_i) - F_{boyi} \cos(\theta_c + \varphi_i)] = -T_{out} \\ \sum_{i=1}^n F_{boxi} - k_o \delta X_o = 0 \\ \sum_{i=1}^n F_{boyi} - k_o \delta Y_o = 0 \end{cases} \quad (60)$$

2.3 Analysis result and discussion

Table 1 details the core design parameters of the RV reducer implemented in this study.

2.3.1 Force analysis results

Figure 13 depicts the force and stress distribution of modified cycloid-pin pairs at 70° and 140° input angles. The contact analysis reveals that meshing tooth counts consistently remain below 50% post-modification. This phenomenon primarily occurs because the profile modification increases the initial clearance between meshing components, thereby requiring the cycloid gear to rotate through a greater angle under torque before establishing contact with the pins. Consequently, this mechanical behavior reduces the number of contacting teeth. Notably, the diminished contact tooth quantity necessitates increased contact forces between the individual pins and cycloid gear to maintain equivalent transmission performance. During the two analyzed instances shown in the Fig. 13, the maximum contact forces of 4021.38N and 4024.83N as well as the maximum stresses of 1770.99 MPa and 1770.44 MPa were observed at Pin 17 and Pin 24, respectively. This demonstrates that tooth profile modification constitutes a critical factor influencing pin contact forces in cycloid gear systems.

Figure 14 illustrates the dynamic load distribution across turning-arm bearings during a complete input angle cycle. The reducer incorporates three crankshafts, each equipped with two turning-arm bearings. For nomenclature clarity, the bearings on the first crankshaft are labeled as 1 and 1*, while those on the second and third crankshafts

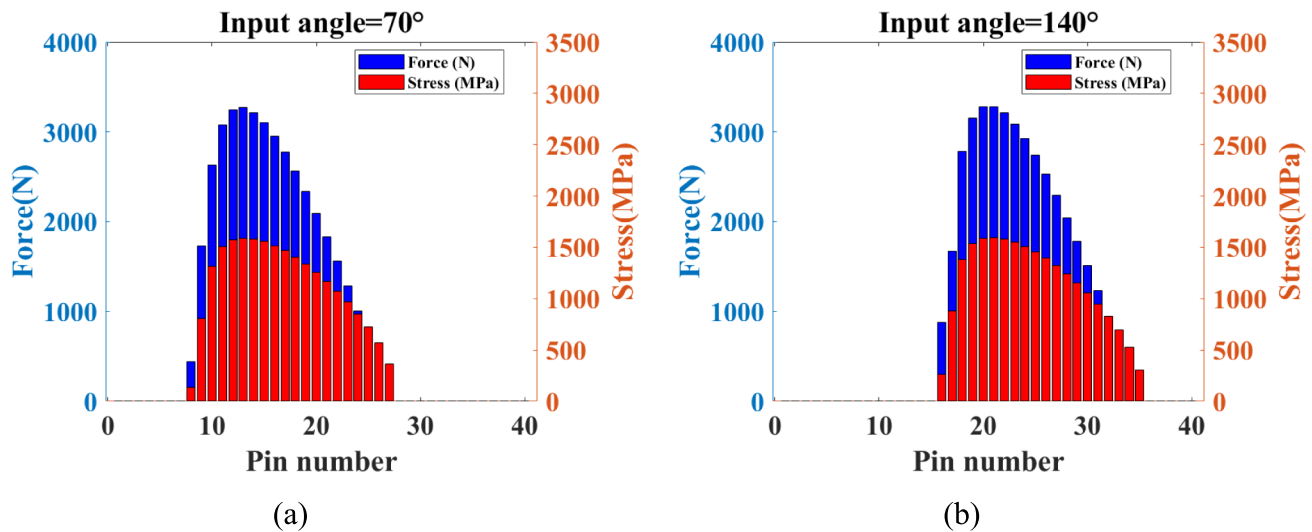


Fig. 13 Contact force and stress of cycloid-pin pairs at input angles of **a** 70° and **b** 140°

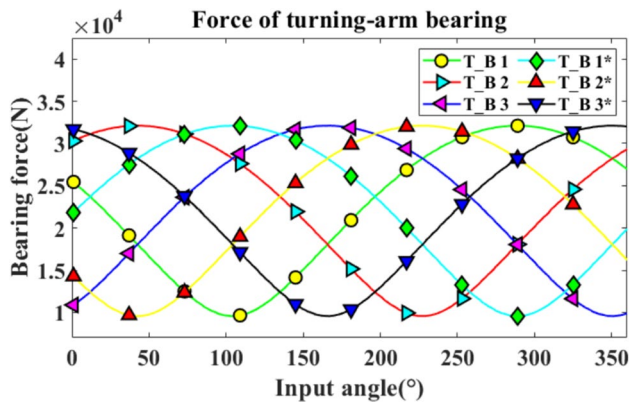


Fig. 14 Force of turning-arm bearings

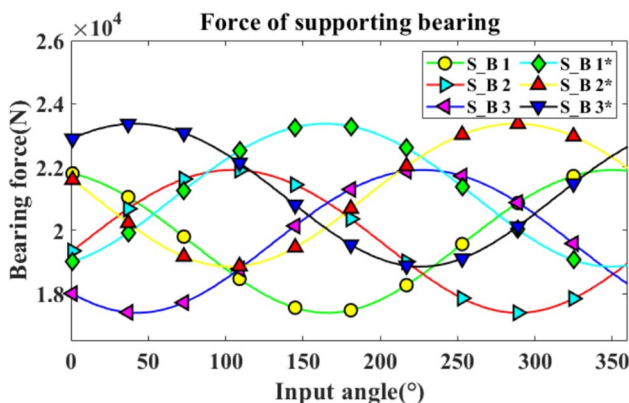


Fig. 15 Force of supporting bearings

are designated as 2/2* and 3/3*, respectively. The loads acting on the bearings demonstrate a pronounced periodic characteristic. Specifically, a phase difference of $2\pi/3$ exists between every two bearing loads on the same cycloid gear, while the two bearings mounted on identical crankshafts

exhibit a half-period phase difference in their load distributions. The bearing forces in magnitude are varying in the range of 450N~34kN.

Figure 15 illustrates the dynamic load distribution across supporting bearings during a complete input angle cycle. For nomenclature clarity, the support bearings numbered 1*, 2*, and 3* are positioned adjacent to the involute gear transmission mechanism, whereas the bearings designated as 1, 2, and 3 are located in proximity to the output mechanism assembly. Similarly, the loads acting on the bearings presents a significant periodic characteristic. Specifically, a phase difference of $2\pi/3$ exists between every two bearing loads on the same section plane, while the two bearings mounted on identical crankshafts exhibit a half-period phase difference in their load distributions. However, the bearing forces adjacent to the involute gear transmission mechanism are varying in the range of 19kN~23kN, larger than those near the output mechanism assembly varying in the range of 17kN~22kN.

2.3.2 Surface wear of cycloid-pin pairs

Figure 16 displays the wear characteristics of cycloid-pin gear pairs after 6000 operational hours. With a predefined threshold of 3 μm , established as the critical wear depth, exceeding this limit necessitates updating the mating components' geometric parameters and reintroducing them into the iteration cycle. The graphical representation reveals that tooth flank wear demonstrates a progressive increase with extended service duration, while simultaneously exhibiting a tendency towards more uniform wear distribution. The wear depth on the cycloid gear is between $4.7 \times 10^{-3}\text{mm}$ and $5.7 \times 10^{-3}\text{mm}$, while the wear depth on the pins is between $5 \times 10^{-3}\text{mm}$ and $6 \times 10^{-3}\text{mm}$. Furthermore, dual

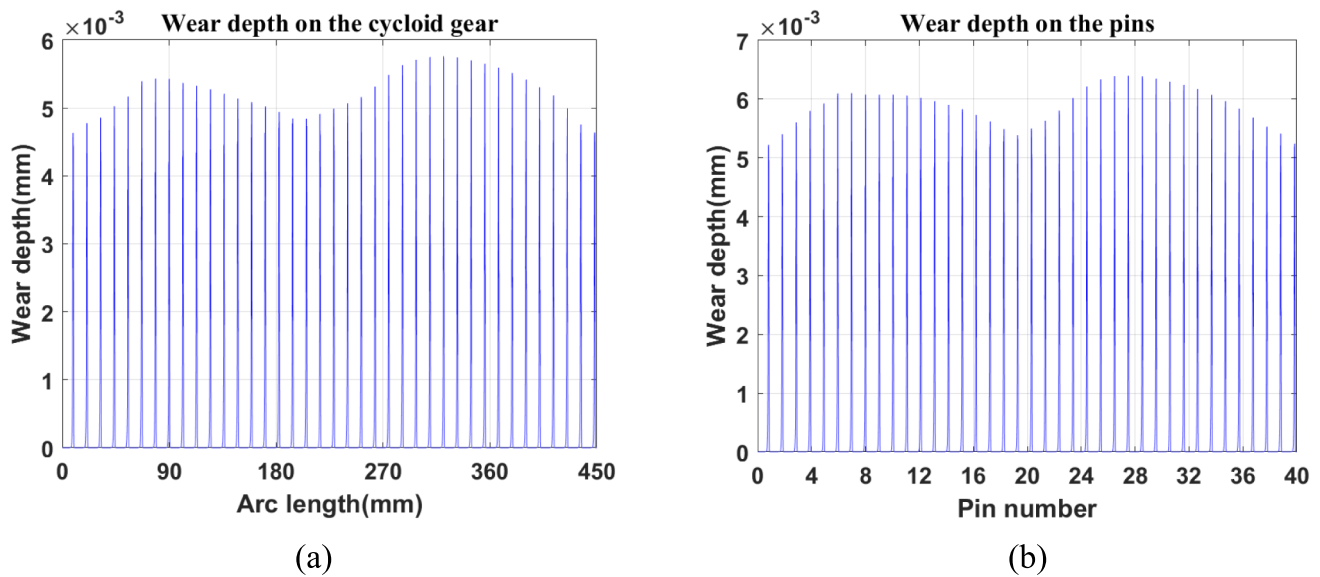


Fig. 16 Wear depth within 6000 h: **a** cycloid gear and **b** pin

comparable wear peaks are identified on both the cycloid disk and pin tooth surfaces. This phenomenon may result from the following mechanism: Once accumulated wear depth on meshing surfaces surpasses the predefined threshold ($3\text{ }\mu\text{m}$), geometric parameters of the gear pair are consequently updated and subsequently reintroduced into the LTCA model for iterative recalibration. This parametric modification leads to a redistribution of contact stress patterns across the tooth surfaces, which not only generates new localized wear peaks but also promotes progressive homogenization of wear distribution through evolutionary adaptation of the meshing behavior.

2.3.3 Effect of single component error on transmission precision

Given that the system transmission error is influenced by deviations in individual components, this section investigates the variation patterns of transmission error under the isolated effect of a single error source, with all other error components held at zero. The phase angle for each error component was fixed at 0° , while their magnitudes were configured into three comparative levels ($5\text{ }\mu\text{m}$, $15\text{ }\mu\text{m}$, and $25\text{ }\mu\text{m}$). Table 2 below enumerates the detailed parameter settings.

Figure 17 illustrates the system transmission precision, quantified as the numerical spread between minimum and maximum transmission errors, under isolated component error effects. Component-induced errors within involute transmission mechanisms, including sun gear assembly, sun gear machining eccentricity error, and planet gear machining eccentricity error, demonstrate negligible impacts on

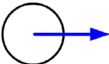
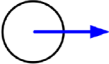
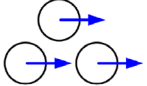
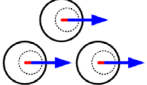
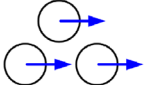
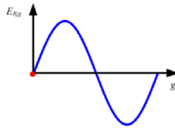
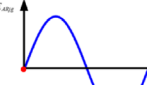
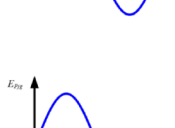
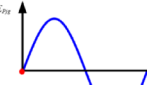
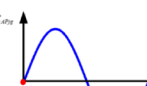
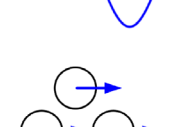
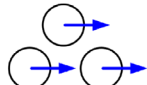
system transmission precision. Even under maximum error magnitudes for these components, the resultant degradation in transmission precision remains limited to $3.5''$. However, component-induced errors detrimentally affect transmission precision in cycloid-pin gear transmissions, notably the eccentricity error of the crank shaft cams. At an error magnitude of $25\text{ }\mu\text{m}$, the resultant transmission precision degradation approaches $80''$. As a result, maintaining stringent control over component geometric errors in cycloid-pin transmission mechanisms is critical to achieving superior system transmission precision.

2.3.4 Effect of surface wear on transmission precision

Given that prolonged operation induces progressive tooth surface wear on cycloid gears, which further influences system transmission precision, this subsection systematically investigates the resultant impact by configuring five distinct levels of tooth flank wear ($10\text{ }\mu\text{m}$, $20\text{ }\mu\text{m}$, $30\text{ }\mu\text{m}$, $40\text{ }\mu\text{m}$, and $50\text{ }\mu\text{m}$) as detailed in the Table 3 below.

Taking the cycloid gear wear case BX5 as an example, Fig. 18a simulates the corresponding transmission error. The transmission error changes very regularly, with a fluctuation range of $-3.9974'' \sim 6.1871''$ and a corresponding transmission precision of $10.1845''$. The other error groups are simulated according to the same steps, and the transmission precision shown in Fig. 18b can be obtained. From Fig. 18, it can be observed that as the wear of the cycloid gear increases from $10\text{ }\mu\text{m}$ to $50\text{ }\mu\text{m}$, the transmission precision increases from $1.4032''$ to $10.1845''$, a relative increase of $8.7813''$.

Table 2 Component error parameter value

Error Name	Error Diagram	Error Value (A_i, α_i)	Item Number
Sun gear assembly error		(5μm, 0°)	SA1
		(15μm, 0°)	SA2
		(25μm, 0°)	SA3
Sun gear machining eccentricity error		(5μm, 0°)	SE1
		(15μm, 0°)	SE2
		(25μm, 0°)	SE3
Planet gears machining eccentricity error		(5μm, 0°)	SP1
		(15μm, 0°)	SP2
		(25μm, 0°)	SP3
Cams on crank shafts machining eccentricity error		(5μm, 0°)	CE1
		(15μm, 0°)	CE2
		(25μm, 0°)	CE3
Crank shaft holes on cycloid gears machining eccentricity error		(5μm, 0°)	HE1
		(15μm, 0°)	HE2
		(25μm, 0°)	HE3
Radial tooth profile deviation of cycloid gears		$E_{R/g} = 5 \sin \frac{2\pi(g-1)}{40}$	RE1
		$E_{R/g} = 15 \sin \frac{2\pi(g-1)}{40}$	RE2
		 $E_{R/g} = 25 \sin \frac{2\pi(g-1)}{40}$	RE3
Accumulated pitch deviation of cycloid gears		$E_{A/R/g} = 5 \sin \frac{2\pi(g-1)}{40}$	AE1
		$E_{A/R/g} = 15 \sin \frac{2\pi(g-1)}{40}$	AE2
		 $E_{A/R/g} = 25 \sin \frac{2\pi(g-1)}{40}$	AE3
Radial tooth profile deviation of the pinwheel		$E_{P/g} = 5 \sin \frac{2\pi(g-1)}{40}$	DE1
		$E_{P/g} = 15 \sin \frac{2\pi(g-1)}{40}$	DE2
		$E_{P/g} = 25 \sin \frac{2\pi(g-1)}{40}$	DE3
Accumulated pitch deviation of the pinwheel		$E_{A/P/g} = 5 \sin \frac{2\pi(g-1)}{40}$	GE1
		$E_{A/P/g} = 15 \sin \frac{2\pi(g-1)}{40}$	GE2
		 $E_{A/P/g} = 25 \sin \frac{2\pi(g-1)}{40}$	GE3
Crank shaft holes on output disc machining eccentricity error		(5μm, 0°)	TE1
		(15μm, 0°)	TE2
		(25μm, 0°)	TE3
Output disc assembly error		(5μm, 0°)	XE1
		(15μm, 0°)	XE2
		(25μm, 0°)	XE3

3 Simulation analysis based on RecurDyn

This section employs RecurDyn for dynamic simulations to validate the aforesaid theoretical model. It should be noted that certain error factors, including radial profile deviations of cycloid gears, accumulated pitch deviations of cycloid gears, radial profile deviations of pinwheel and accumulated pitch deviations of pinwheel, are temporarily excluded from consideration due to the inherent challenges in simulating these parameters through the model. Consequently, this study focuses on the impact of component errors in the sun gear, planetary gears, cycloid gear, crankshaft, and output disc on transmission precision. Each error is set to 10 μm at 0° phase angle, using the RV reducer parameters in Table 1.

As illustrated in Fig. 19, these diagrams respectively present the schematic representations of bearing forces on the input planetary carrier, the two cycloid gears, and the output planetary carrier. For clarity, the figures display only the simulation data within the time range of 0 to 4 s. Based on the bearing force results from these figures, a comparison with the theoretical values from Sect. 2.3.1 is provided in Table 4.

As shown in Table 4, a comparison of the simulated and theoretical bearing forces acting on the input planetary carrier, output planetary carrier, cycloid gear 1, and cycloid gear 2 reveals minor discrepancies between simulation results and theoretical values. The maximum relative error is merely 8.89%, which not only validates the simulation model's validity but also confirms the reasonableness of the theoretical force analysis model for RV reducer components established earlier.

As observed in Fig. 20 the transmission error range derived from software simulation spans from -30.69" to 27.69" with a transmission precision of 58.55", while the theoretical model yields a range of -29.56" to 26.29" and an accuracy of 55.85", indicating minimal discrepancy between the two approaches. Furthermore, regarding error distribution patterns, although the waveforms of transmission errors exhibit partial misalignment in localized regions, they demonstrate substantial similarity in overall morphology. Both distributions exhibit approximately symmetric distribution about the zero point of the vertical axis, characterized by significant positive and negative fluctuations and discernible periodicity—features consistent with actual operating behavior of RV reducers. These findings validate the correctness of the transmission error mathematical model for RV reducers established.

To quantify the influence of tooth flank wear depth on transmission error, uniform isometric modification was applied to cycloid gear profiles during SolidWorks modeling to approximate equivalent wear depths. Dynamic

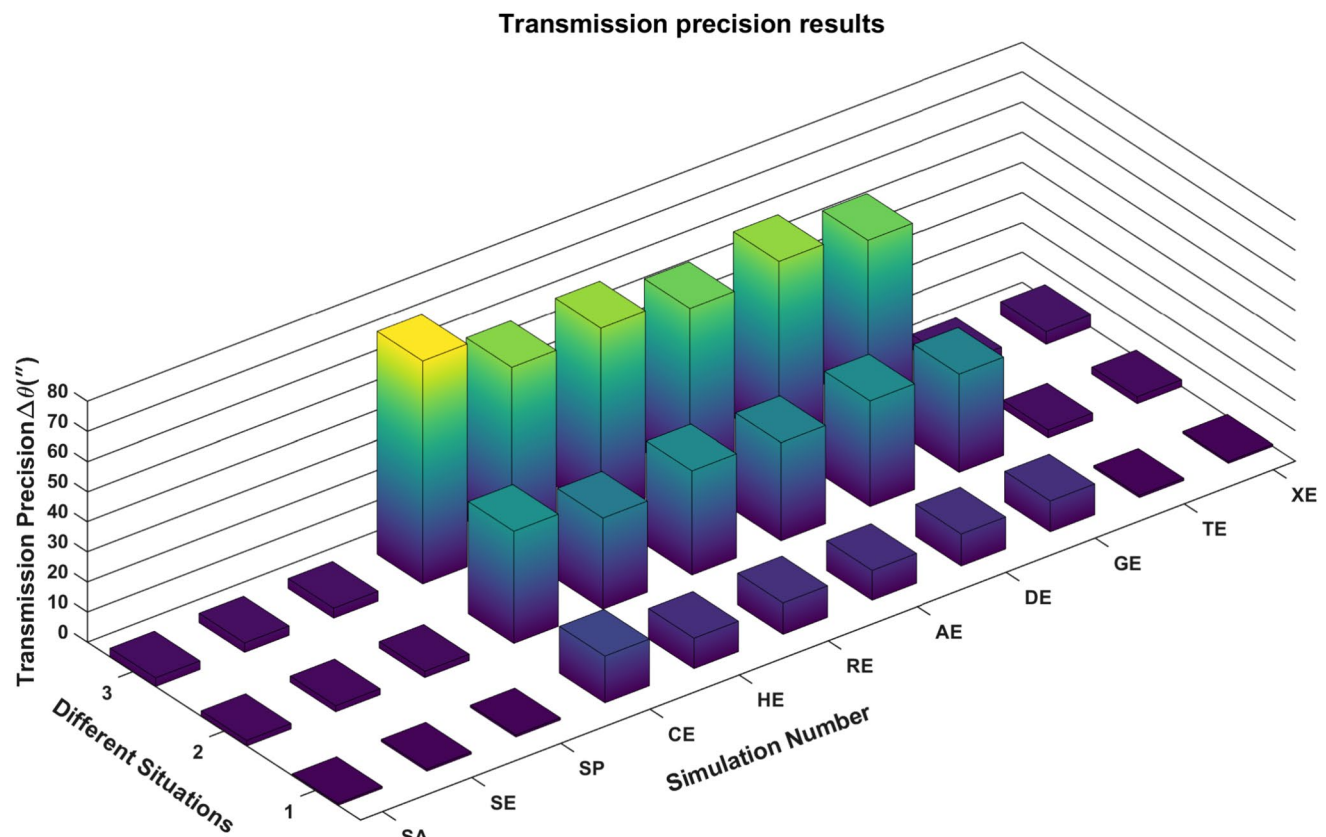


Fig. 17 Transmission precision results

Table 3 Tooth surface wear of cycloid gears

Item name	Wear magnitude (μm)	Item number
Surface wear	10	BX1
	20	BX2
	30	BX3
	40	BX4
	50	BX5

simulations in RecurDyn were performed for three configured wear depth groups: 15 μm , 25 μm , and 35 μm .

Figure 21a presents the transmission error derived from RecurDyn simulations for a cycloid gear tooth wear depth of 15 μm . Correspondingly, Fig. 21b displays the simulated error obtained via the theoretical model by incorporating component errors and wear depth parameters of the simulation model. Similarly, Fig. 22 respectively show transmission errors for 25 μm wear depth acquired through RecurDyn simulations and theoretical modeling. Figure 23 further provide transmission error results for 35 μm wear depth from software simulations and theoretical solutions. From the Figs, it can be obtained that comparative analysis of simulation and theoretical results under tooth wear depths of 15 μm , 25 μm , and 35 μm reveals that:

- (1) Transmission errors from both approaches exhibit comparable fluctuation ranges.
- (2) The maximum relative error in transmission precision is merely 6.31%, falling within acceptable tolerance limits.

These findings demonstrate the feasibility of rigid-body dynamics simulations for RV reducers in RecurDyn, while validating the correctness of the quasi-static mechanical analysis model established in this study.

4 Transmission performance test

The experimental prototype was designed based on the specifications of a common industrial RV reducer model 320E. The experiment aims to assess the RV reducer's transmission error, lost motion, and efficiency. This requires separate measurements of the input/output rotational angles and torque values from the prototype. The experimental data are transmitted via a data acquisition card to an industrial computer for centralized processing, enabling subsequent derivation of the aforementioned performance parameters through data analysis. Accordingly, several performance testing platforms for RV reducers was developed to fulfill

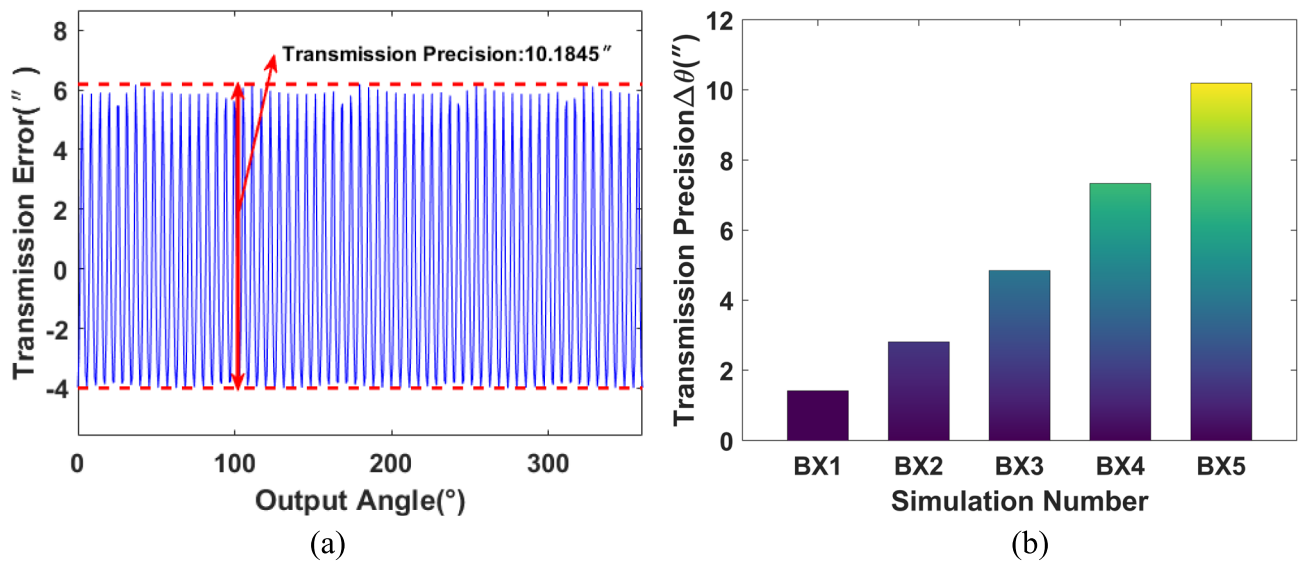


Fig. 18 Transmission error results: **a** result of BX5 and **b** reference comparison chart

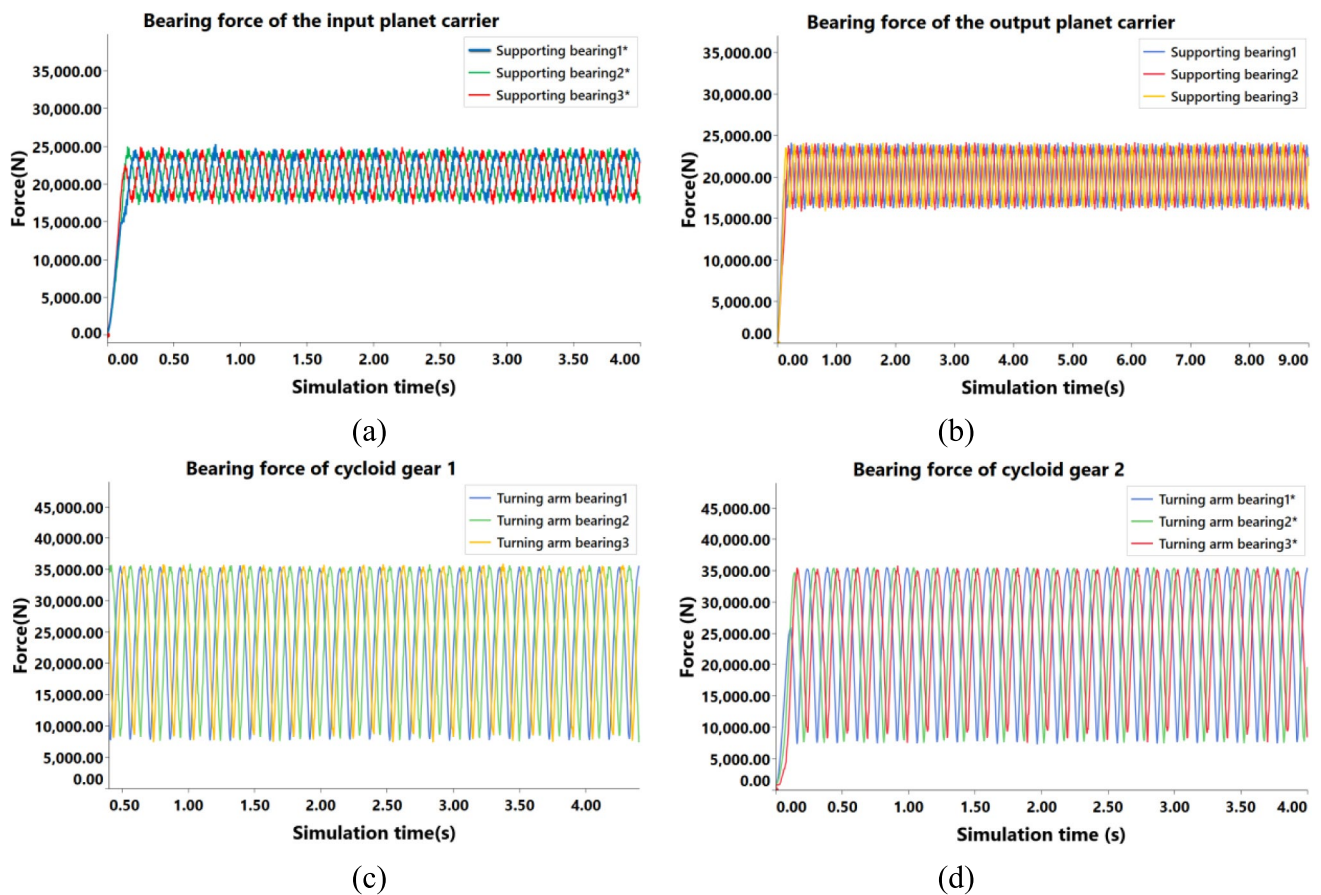
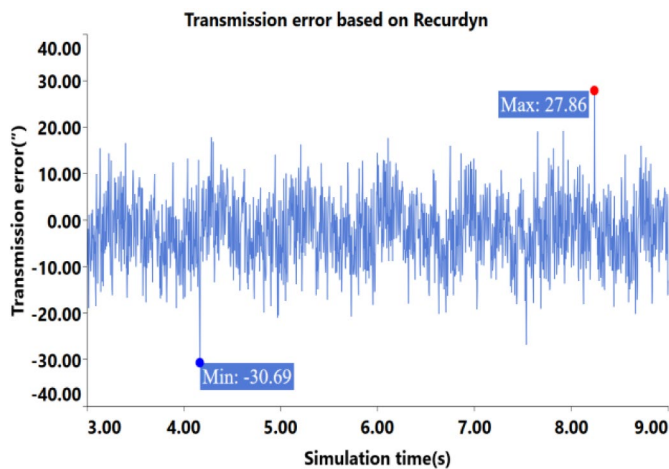


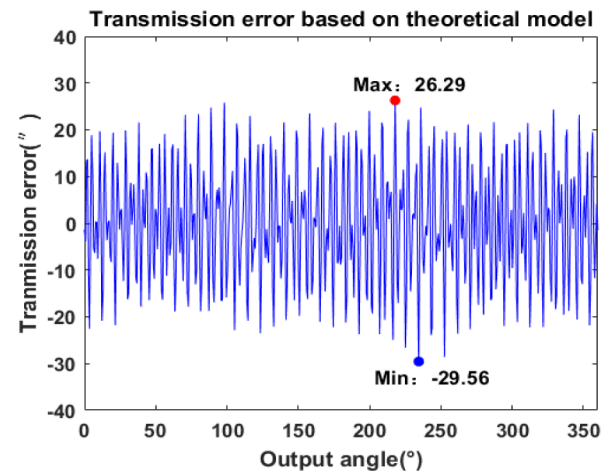
Fig. 19 Bearing force results based on dynamic simulation: **a** bearing forces of the input planet carrier, **b** bearing forces of the output planet carrier, **c** bearing forces of the cycloid gear 1 and **d** bearing forces of the cycloid gear 2

Table 4 Comparison between simulation and theoretical results of bearing forces

Item name			Simulation value (N)	Theoretical value (N)	Relative error (%)
Input planet carrier	Supporting Bearing 1*	Max	25,274	23,682	6.72
		Min	17,170	18,540	-7.39
	Supporting Bearing 2*	Max	24,890	23,723	4.92
		Min	17,219	18,543	-7.14
	Supporting Bearing 3*	Max	24,963	23,677	5.43
		Min	17,266	18,538	-6.86
Output planet carrier	Supporting Bearing 1	Max	24,136	22,224	8.60
		Min	16,052	17,084	-6.04
	Supporting Bearing 2	Max	24,195	22,219	8.89
		Min	15,852	17,037	-6.96
	Supporting Bearing 3	Max	24,165	22,223	8.74
		Min	15,884	17,082	-7.01
Cycloid gear 1	Turning arm Bearing 1	Max	35,631	33,904	5.09
		Min	7602	8033	-5.37
	Turning arm Bearing 2	Max	35,923	33,687	6.64
		Min	7506	7986	-6.01
	Turning arm Bearing 3	Max	35,953	33,685	6.73
		Min	7391	7976	-7.33
Cycloid gear 2	Turning arm Bearing 1*	Max	35,641	33,675	5.84
		Min	7305	7766	-5.94
	Turning arm Bearing 2*	Max	35,711	33,693	5.99
		Min	7445	8005	-7.00
	Turning arm Bearing 3*	Max	35,796	33,695	6.24
		Min	7518	8020	-6.26



(a)



(b)

Fig. 20 Transmission error results: **a** based on Recurdyn and **b** based on theoretical model

these measurement requirements. Figure 24 shows a compositional diagram of the comprehensive performance test platform for RV reducers.

Subsequently, the corresponding experimental RV reducer prototype for testing was machined and fabricated, as illustrated in Fig. 25.

Based on measurements obtained via coordinate measuring machine (CMM), the experimental prototype exhibited a radial profile deviation of 4 μm and an accumulated pitch deviation of 3 μm for the cycloid gear, while the pinwheel showed corresponding deviations of 5 μm and 3 μm , respectively; the main error parameters of all other components are documented in Table 5.

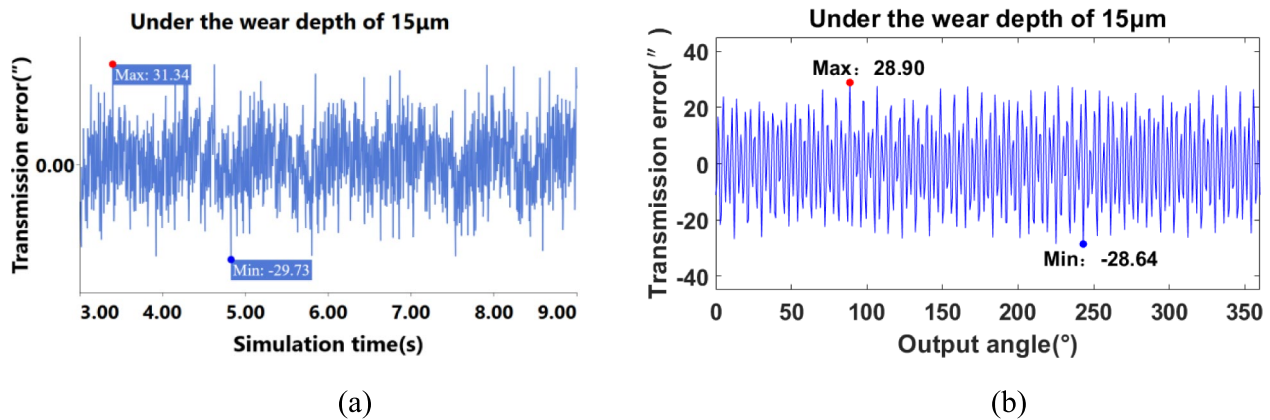


Fig. 21 Transmission error under 15 μm: **a** RecurDyn and **b** theoretical model

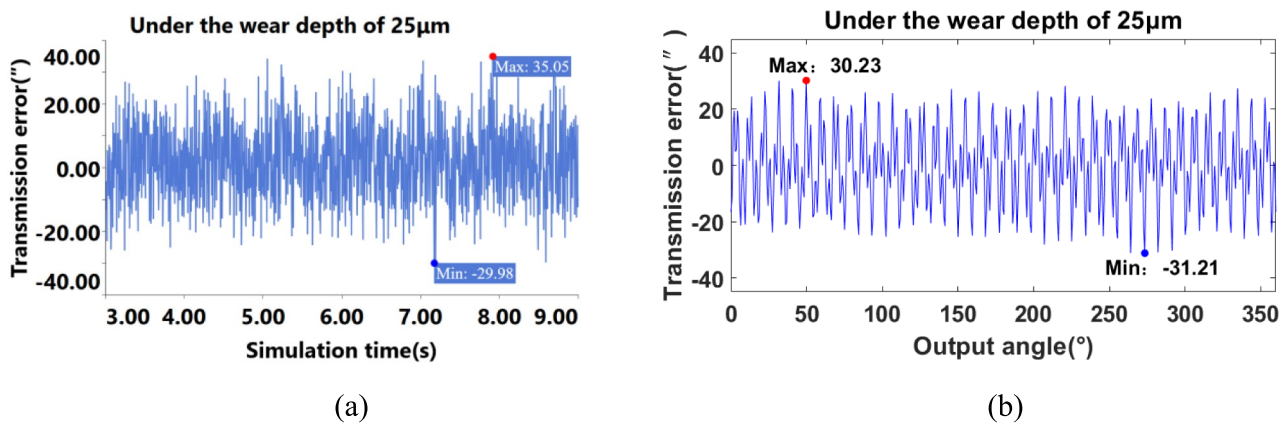


Fig. 22 Transmission error under 25 μm: **a** RecurDyn and **b** theoretical model

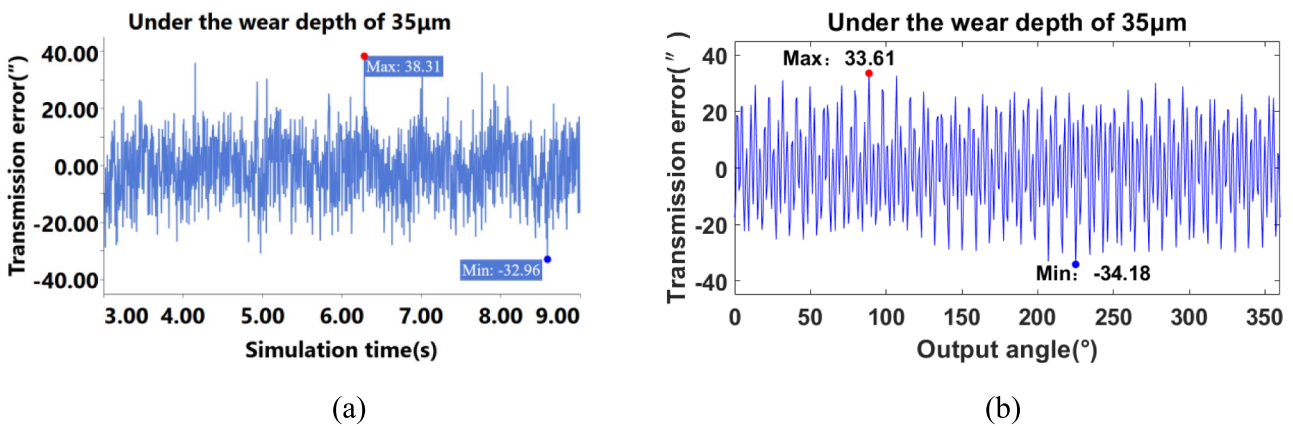


Fig. 23 Transmission error under 35 μm: **a** RecurDyn and **b** theoretical model

To ensure the validity of comparisons between the model and experiments, the phase angles of all geometric errors (including those listed in Table 5 for the sun gear and output disc assembly) were precisely measured directly from the physical prototype prior to testing. A rigorous process was followed to maintain consistency with the coordinate system of the theoretical model. A unified angular reference

baseline was established for both the model and physical components. Using a high-precision coordinate measuring machine, point cloud data for each component's critical features was captured in the final assembled state. Through optimal alignment fitting, the displacement vector of each component's actual position relative to its ideal theoretical position was calculated. The magnitude of this vector

corresponds to the error amplitude, while its direction relative to the preset 0° reference defines the error phase angle. These independently measured error parameters were directly input as known conditions into the theoretical model. This approach ensures that error excitation sources in simulation statistically represent the actual manufacturing and assembly conditions of the physical prototype, thereby maximizing the objectivity and comparability of the verification process.

By incorporating the error parameters from Table 5 into the previously established theoretical model, the simulated transmission error curve of the prototype under multi-error coupling effects was generated, as shown in Fig. 26a. The simulated no-load transmission error curve exhibits significant fluctuations. This characteristic waveform directly reflects the sparse and dynamically changing contact state under no-load conditions. With no significant external torque applied, elastic deformation is negligible, and meshing exhibits high selectivity. Contact occurs only when the combined equivalent initial clearance of the cycloidal pin pairs is extremely small or negative. This instantaneous clearance is jointly determined by the vector sum of profile correction values and the projection of all relevant geometric errors onto the line of action. Consequently, the number of engaged teeth sharply decreases and distributes unevenly between pins. As the input shaft rotates, dominant contact pairs abruptly shift to specific positions according to the error phase distribution, causing step changes in the output angle. This generates observable peaks and valleys within the gear transmission curve. Figure 26b corresponds to the experimentally measured transmission error curve obtained under no-load conditions with an input rotational speed of 750 r/min.

Comparative analysis shows: the theoretical error of the prototype fluctuated between $-11.76''$ and $11.56''$ with a transmission precision of $23.32''$, whereas the experimentally

measured error ranged from $-12.58''$ to $12.90''$ yielding a transmission precision of $25.48''$. The relative deviation between the theoretical and experimental precision values is 9.26%, which falls within the acceptable tolerance (typically $<10\%$) for precision mechanical systems. The observed quantitative discrepancies are attributed to unavoidable experimental factors, including variations in the lubrication state of reducer components, vibrational interference from the test platform, and thermal expansion-induced changes in gear mesh clearances. Despite these differences, the error bandwidths remain statistically comparable (within 9.26%). Moreover, both the theoretical and experimental error profiles exhibit fundamentally consistent periodic evolution patterns with identical phase characteristics.

This harmonic correspondence, confirmed through both visual inspection and statistical analysis, validates that the developed mathematical model qualitatively captures the essential dynamics of transmission error in RV reducers under operational conditions. It thereby demonstrates the model's capacity to simulate the kinematic error mechanisms inherent in high-precision cycloid drive systems.

Figure 27a~d present the transmission precision and error of the RV reducer under no-load conditions at discrete input rotational speeds. Analysis reveals prototype transmission precision of $24.38''$, $23.02''$, $25.48''$, and $26.08''$ at respective input speeds, all maintained within $30''$. As rotational speed increases from 250r/min to 1000r/min, the transmission error waveforms exhibit no fundamental morphological changes, preserving distinct periodic fluctuations relative to input angular displacement. Concurrently, transmission precision demonstrates negligible variation with a marginally increasing trend ($\Delta=3.8''$ over 750r/min range), statistically confirming that input rotational speed exerts minimal influence on both transmission error characteristics and accuracy metrics.

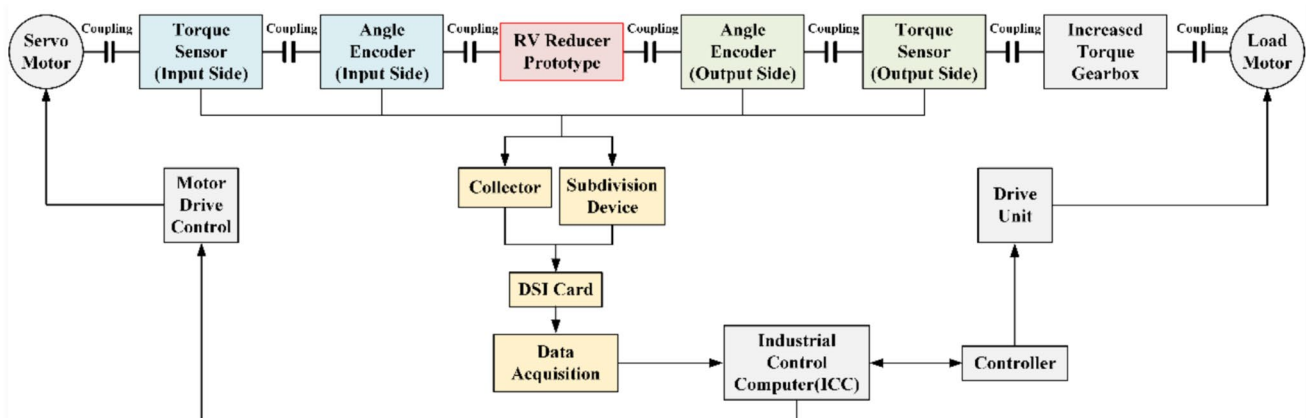


Fig. 24 The compositional diagram of RV reducer comprehensive performance test platform

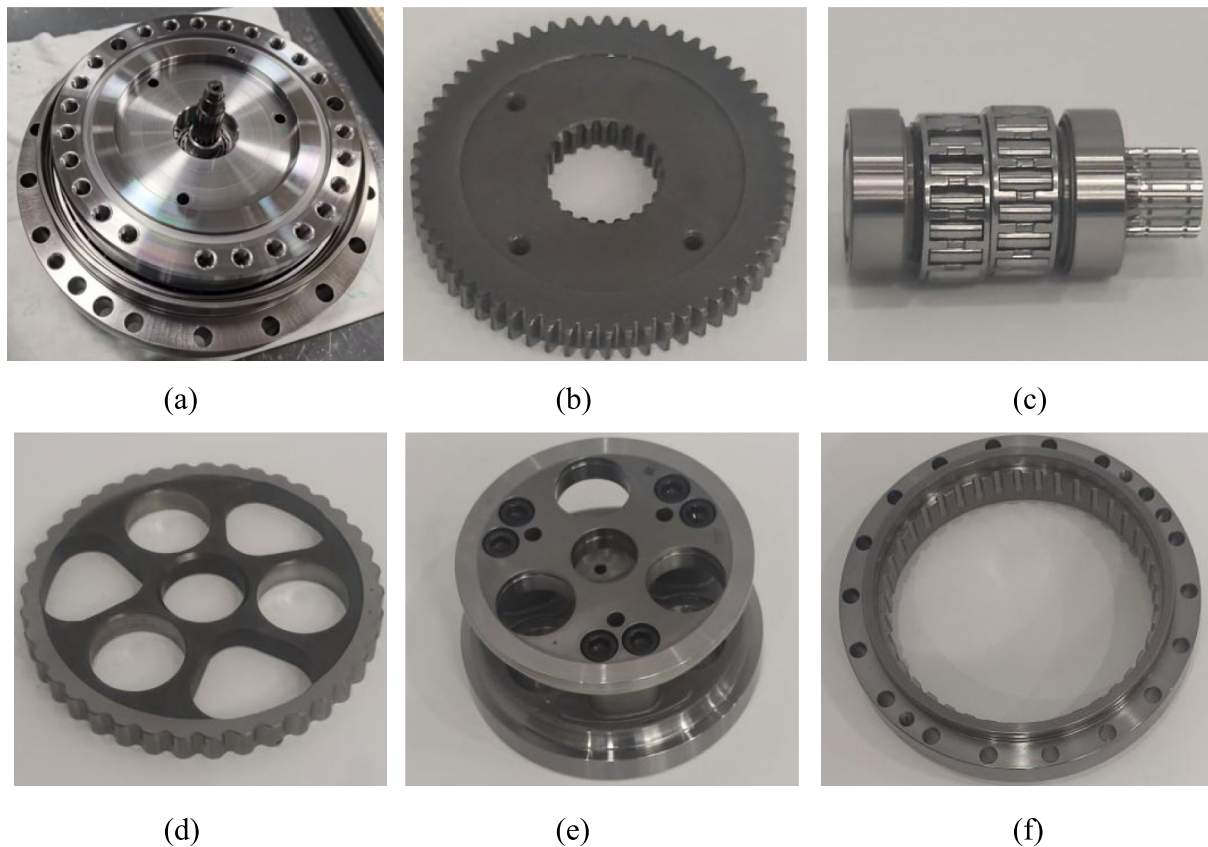


Fig. 25 Main components of the RV reducer: **a** assembled prototype; **b** planetary gear; **c** crankshaft; **d** cycloid gear; **e** planetary carrier assembly; and **f** pin gear

Table 5 Error parameters of main components

Error item	Error value	
	Close to the input side	Close to the output side
Sun gear assembly error	(2 μm , 35°)	-
Sun gear machining eccentricity error	(2.5 μm , 170°)	-
Planet gears machining eccentricity error	(2.5 μm , 120°)	-
	(3 μm , 200°)	-
	(3.5 μm , 80°)	-
Cams on crank shafts machining eccentricity error	(2 μm , 195°)	(3 μm , 55°)
	(2.5 μm , 160°)	(1.5 μm , 345°)
	(1.5 μm , 220°)	(2 μm , 15°)
Crank shaft holes on cycloid gears machining eccentricity error	(3 μm , 65°)	(1.5 μm , 125°)
	(2.5 μm , 80°)	(3 μm , 225°)
	(2 μm , 170°)	(1.5 μm , 305°)
Crank shaft holes on output disc machining eccentricity error	-	(3 μm , 230°)
	-	(2.5 μm , 350°)
	-	(3 μm , 95°)
Output disc assembly error	-	(2.5 μm , 75°)

Transmission efficiency is a crucial performance indicator for RV reducers, and its improvement reduces energy consumption. Figure 28 presents the transmission efficiency under external loads of 0–6000Nm at different input speeds.

The load torque test extends to 6000 Nm, aiming to investigate efficiency trends beyond the typical rated torque range to validate its overload performance margin. Partial performance data of the experimental prototype is shown in Table 6. Overall, when load torque ranges from 0–1500Nm, transmission efficiency increases substantially with rising torque while decreasing noticeably with increasing rotational speed. When load torque spans 1500–6000Nm, transmission efficiency continues to increase across all speeds, but with reduced growth rates and decelerating increments. Concurrently, mutual efficiency differentials narrow rapidly. At the maximum load of 6000Nm, all efficiencies exceed 90%, peaking at approximately 95.42% for 1000r/min.

5 Conclusions

In conclusion, this work investigated the wear behavior, transmission performance, and error sensitivity of RV reducers through an integrated approach integrating theoretical modeling, dynamic simulation, and experimental validation. A wear depth prediction model for cycloidal pins based on the Archard formula was developed, coupled with a quasi-static model for the RV reducer using a mass-spring equivalent model. These

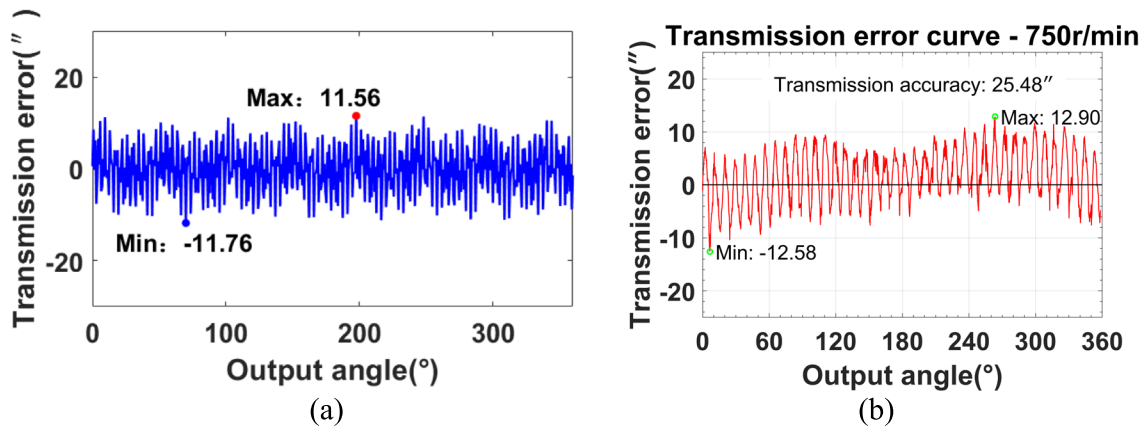


Fig. 26 Transmission error results: **a** simulated results; **b** experimental results

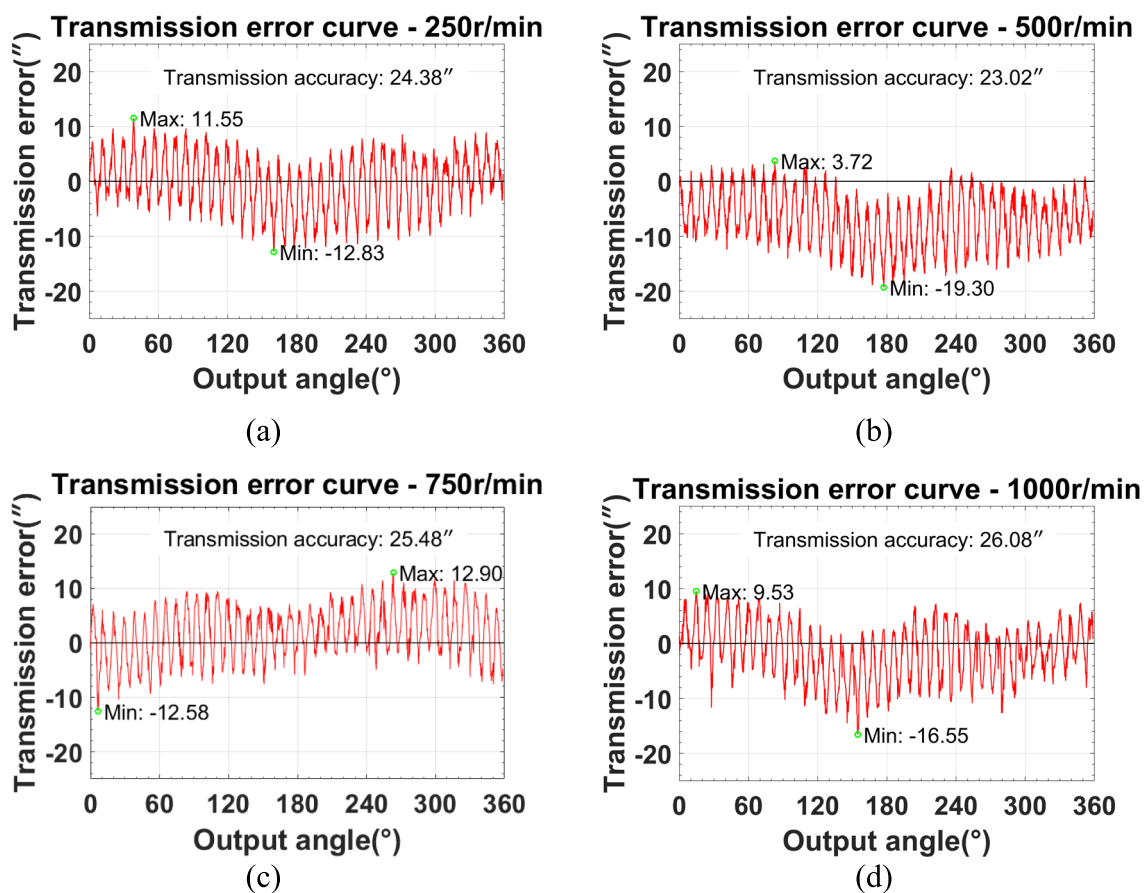


Fig. 27 Transmission error under varying input speeds: **a** 250r/min; **b** 500r/min; **c** 750r/min; and **d** 1000r/min

were supplemented by dynamic analysis using RecurDyn and prototype testing to comprehensively elucidate the intrinsic mechanisms underlying precision degradation, wear evolution, and performance characteristics in RV reducers.

Theoretical modeling demonstrated that the progressive wear process on the tooth flanks of the cycloidal-pin gear pair was precisely analyzed by integrating a wear prediction framework. This framework integrates TCA, LTCA,

and a mesh-based wear evolution algorithm. While tooth profile modification is crucial for load optimization, it reduces the number of teeth in simultaneous contact. This reduction forces individual pins to bear higher contact forces, consequently increasing their susceptibility to wear. Error sensitivity analysis revealed that errors in the cycloidal transmission stage, particularly cam eccentricity error, are the dominant factors underlying precision degradation.

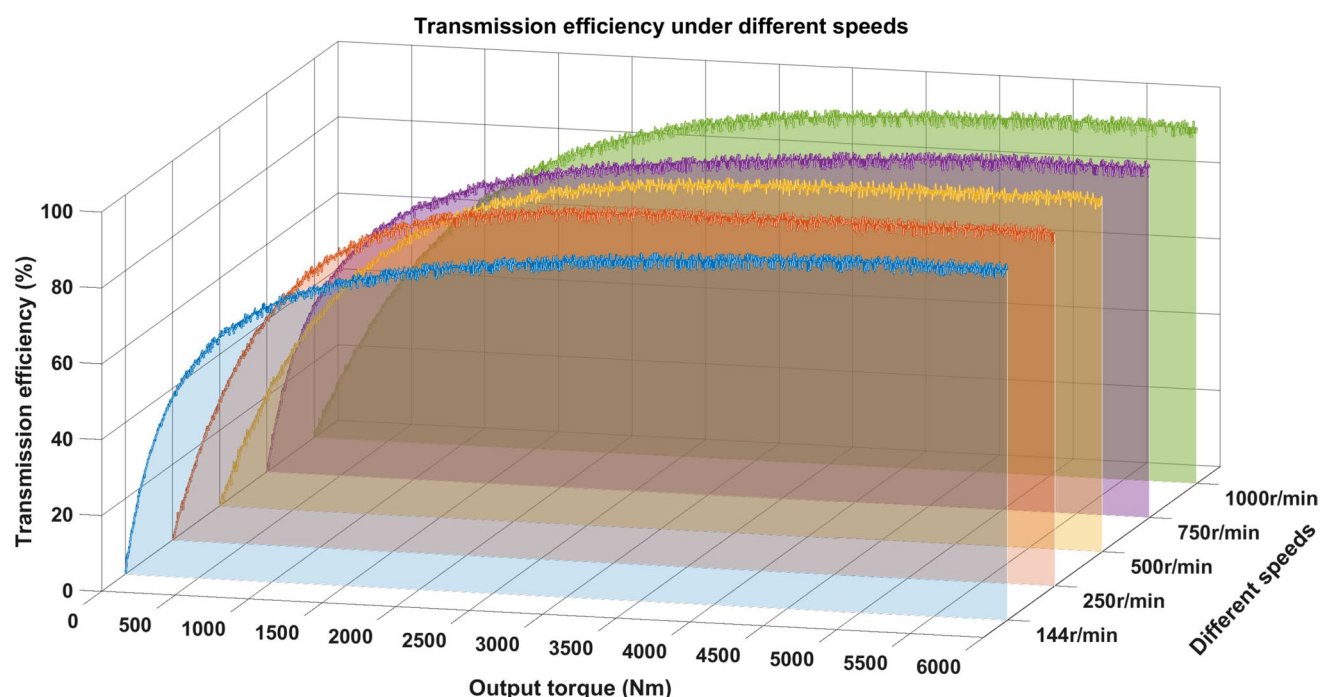


Fig. 28 Transmission efficiency under different speeds

Table 6 Partial performance data for the 320E experimental prototype

Type	Performance data					
Output speed (r/min)	5	10	15	20	25	30
Input power (kW)	3.04	4.94	6.57	8.05	9.41	10.7
Output torque (Nm)	4361	3538	3136	2881	2695	2548

Conversely, errors in the planetary transmission stage exerted minimal impact. Over time, accumulated wear on the cycloidal gear tooth flanks exhibited direct correlation with transmission accuracy loss, confirming wear as the principal degradation mechanism.

Dynamic simulations and prototype testing demonstrated that excellent agreement was observed between simulated and theoretical results for bearing force distributions (max relative error approximately 8.89%) and transmission error characteristics under both rated and worn conditions (max precision error approximately 6.31%). Transmission error measurements under no-load conditions (750 r/min) closely matched model predictions in fluctuation range, periodicity, and precision magnitude (Theoretical: 23.32", Experimental: 25.5"). The prototype demonstrated high efficiency (>90%), reaching a peak value of 95.42% at 1000 r/min and 6000 Nm, with measured backlash of 0.492', further validating the model's practical applicability. Transmission precision showed minimal sensitivity to input speed variations ($\Delta \approx 3.8''$ over 250–1000 r/min). Experimental results quantified key performance metrics: transmission efficiency exhibited strong dependence on both load and speed in the lower torque range (<1500 Nm), increasing with load but

decreasing with speed, before converging at higher loads (>1500 Nm) where efficiencies consistently exceed 90%.

These findings establish a foundation for precision-focused design and maintenance strategies for RV reducers. They facilitate the optimization of tooth profile modification parameters and the design of fault-tolerant structures to mitigate precision degradation caused by wear. Furthermore, quantifiable performance metrics, such as efficiency variation trends across torque ranges, inform optimized motor power matching and integrated lifecycle management.

Acknowledgements This work is supported by National Natural Science Foundation of China (Grants No. 52005354).

Funding National Natural Science Foundation of China, 52005354, Xuan Li.

Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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